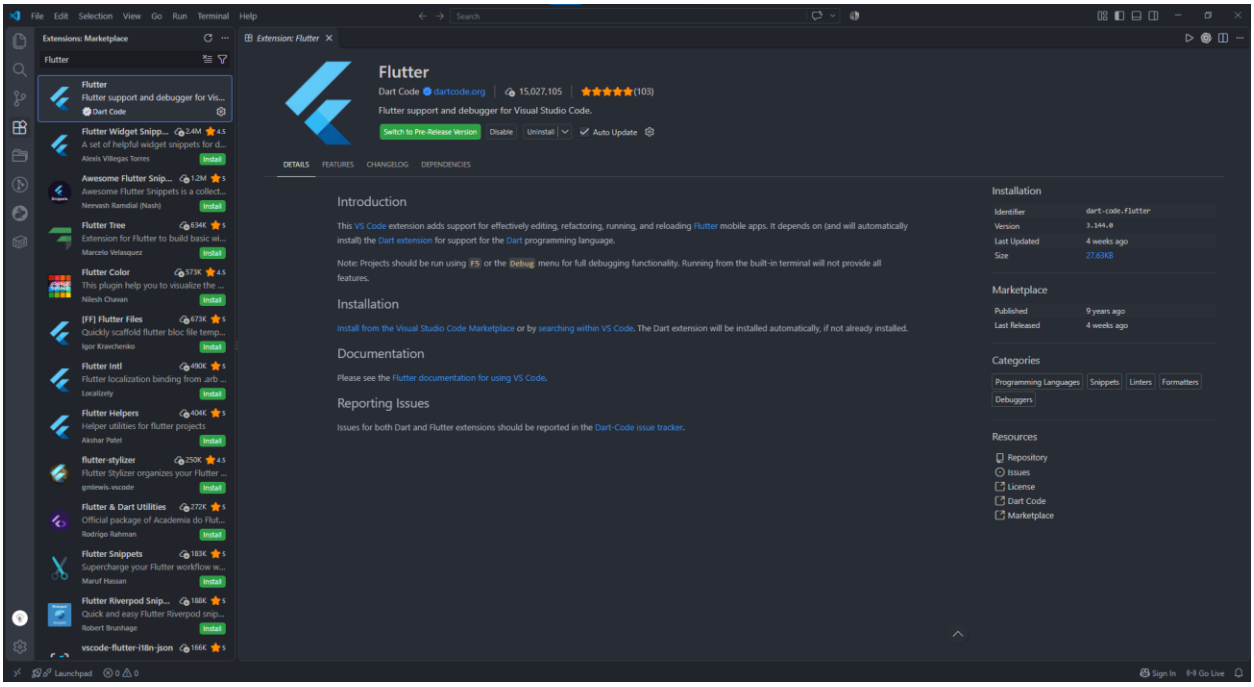
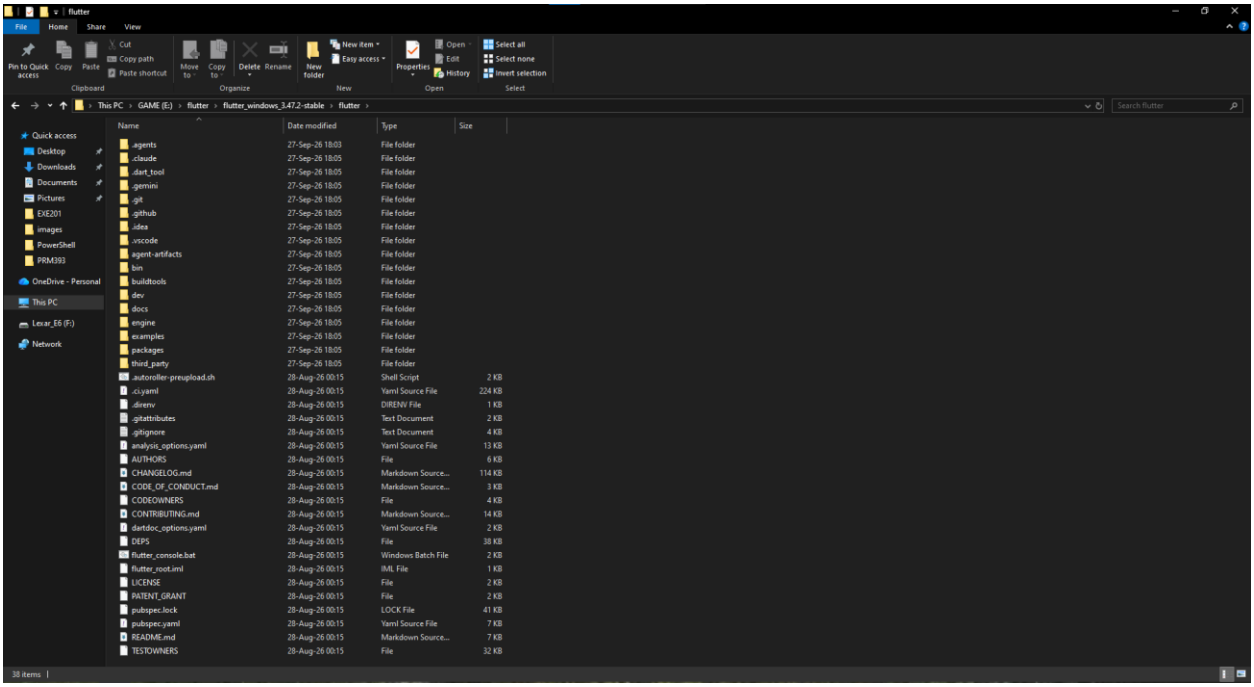
Flutter Course Evidence

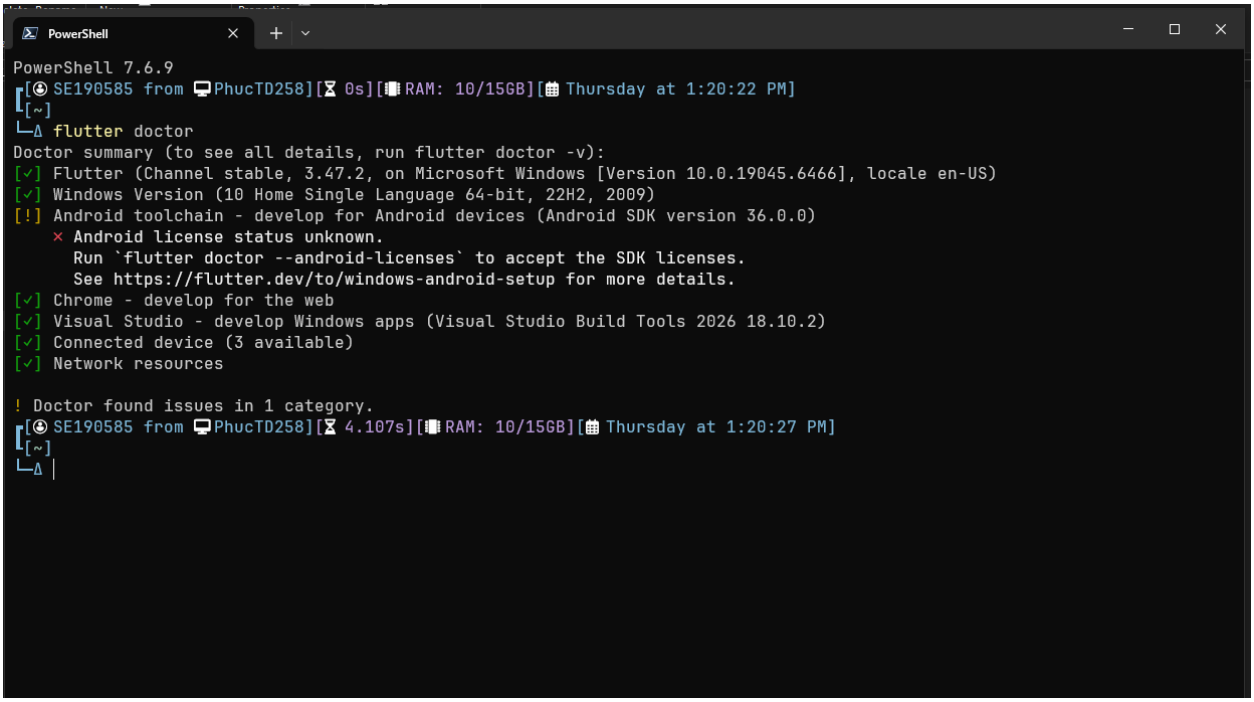
Thông tin sinh viên

Họ và tên	Trần Đình Phúc
MSSV	SE190585
Mã lớp	SE1922
Môn học	PRM393
Khóa học	Flutter Learning Pathway https://docs.flutter.dev/learn/pathway

0/ Install and Set Up Flutter

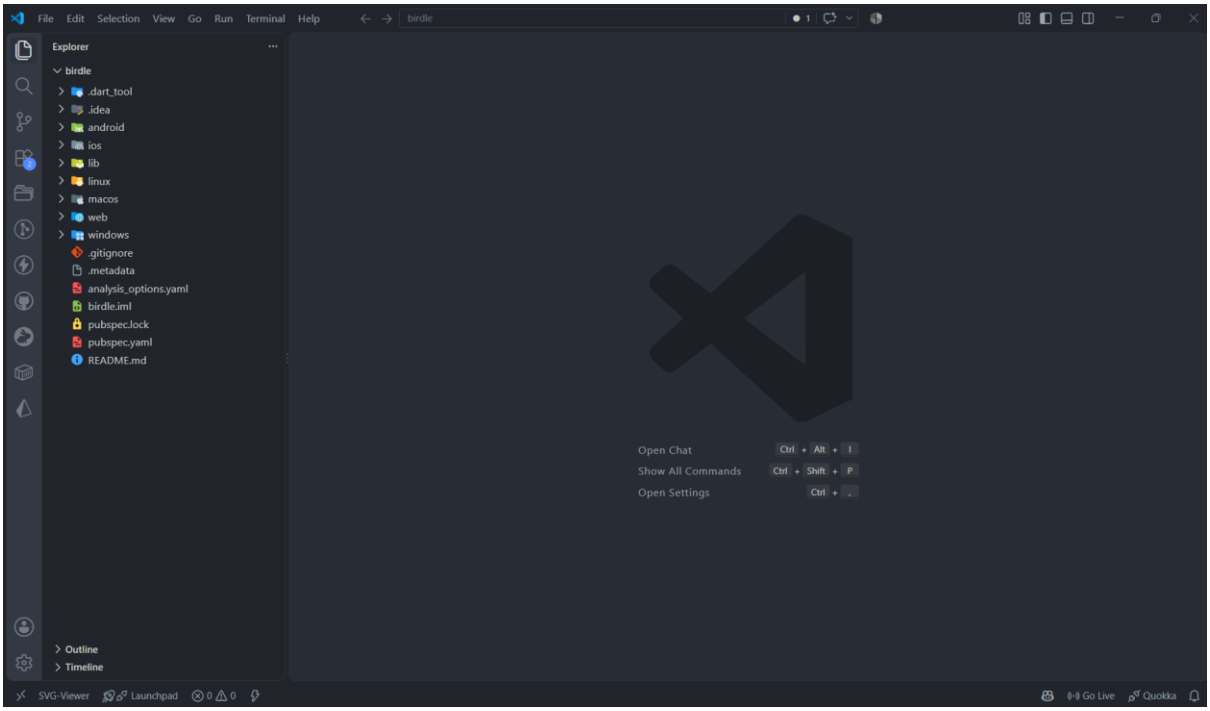




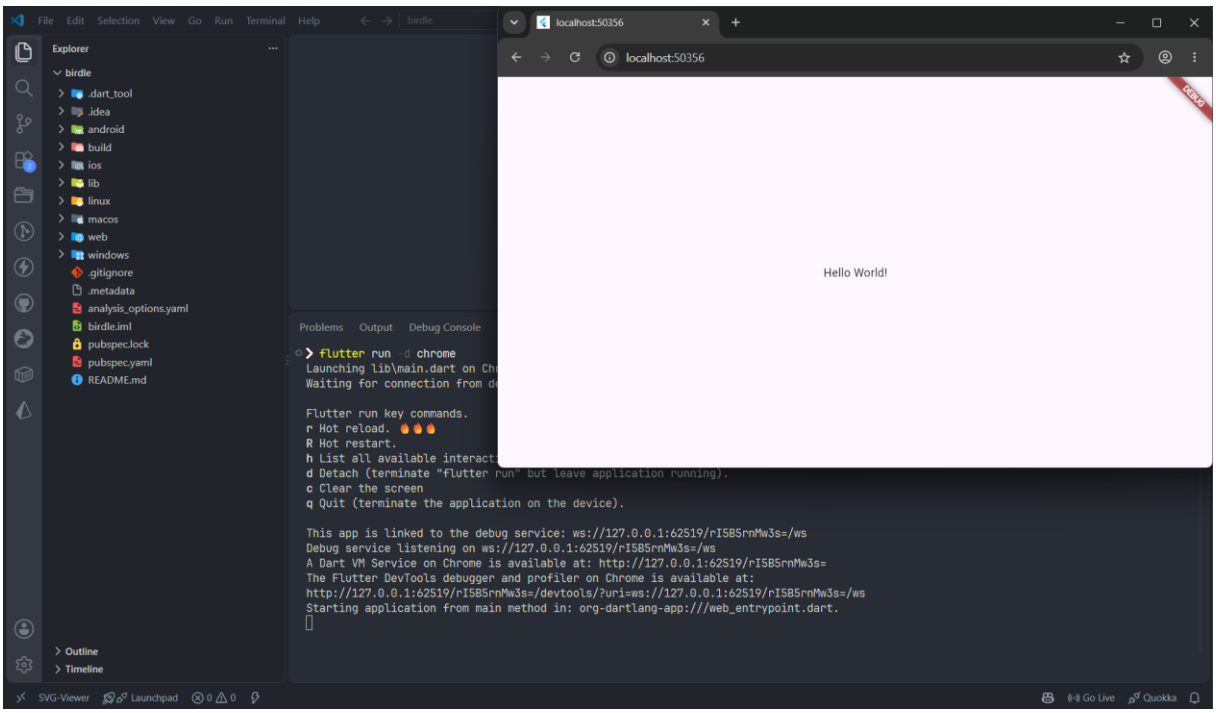


3/ Create a Flutter app

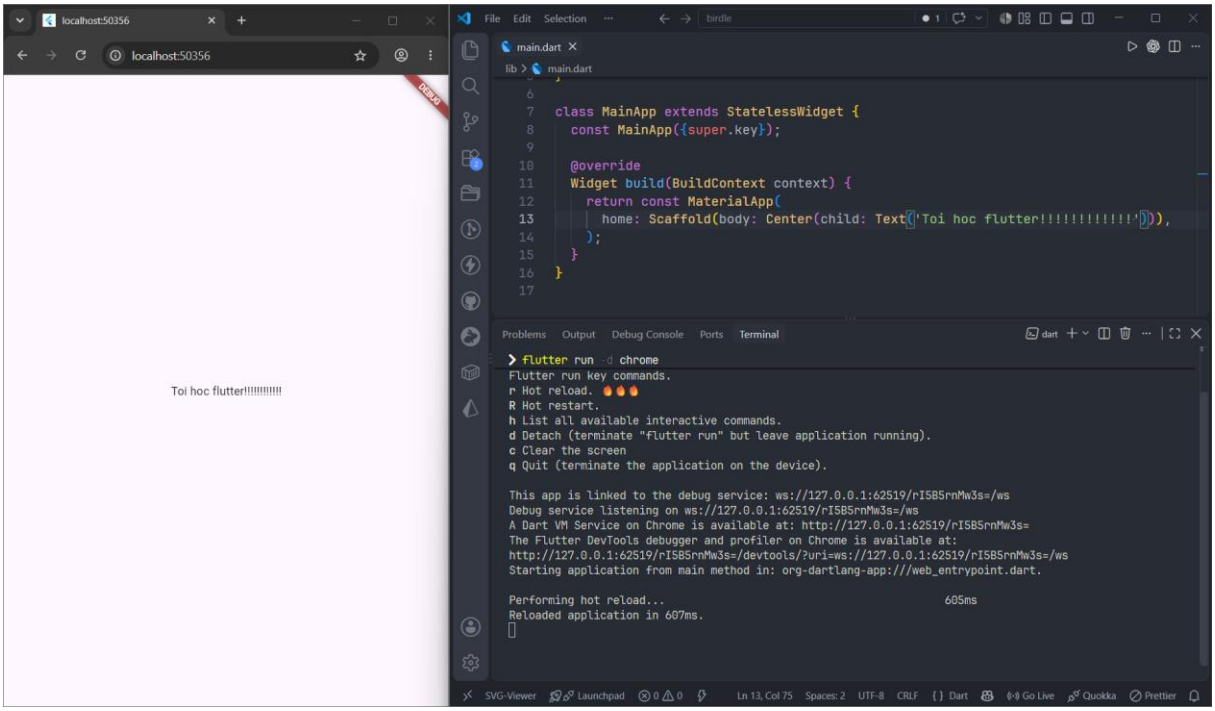
App



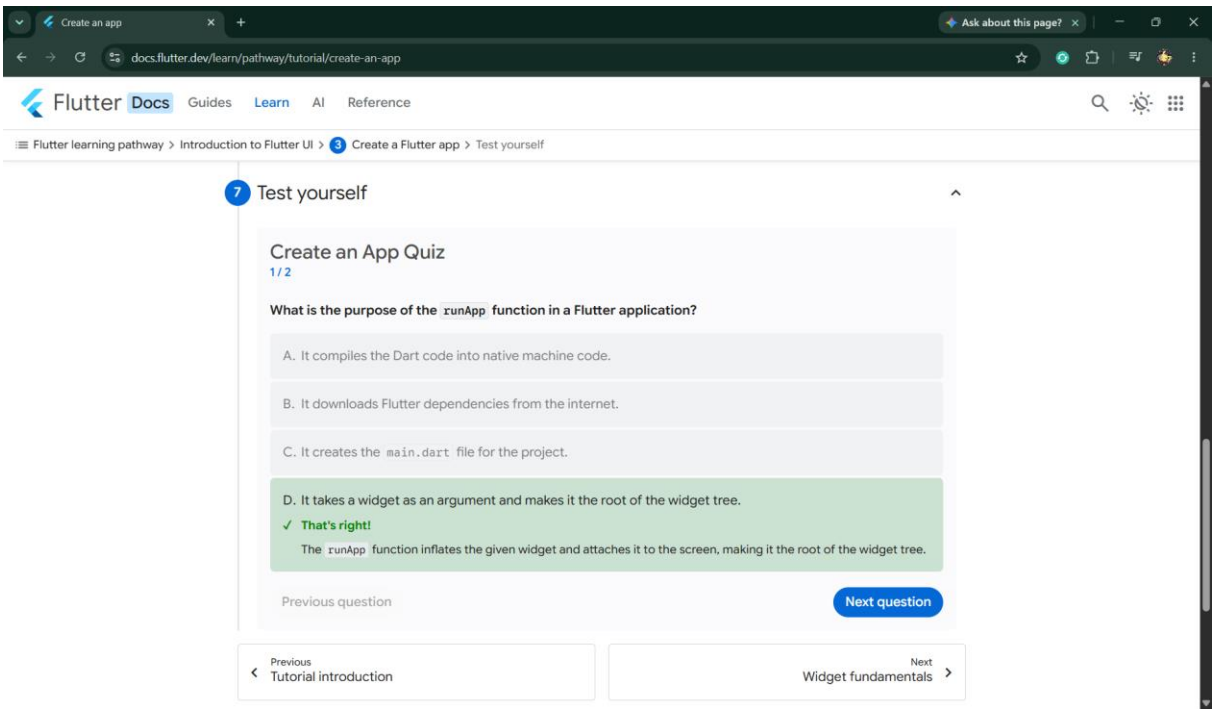
Run app

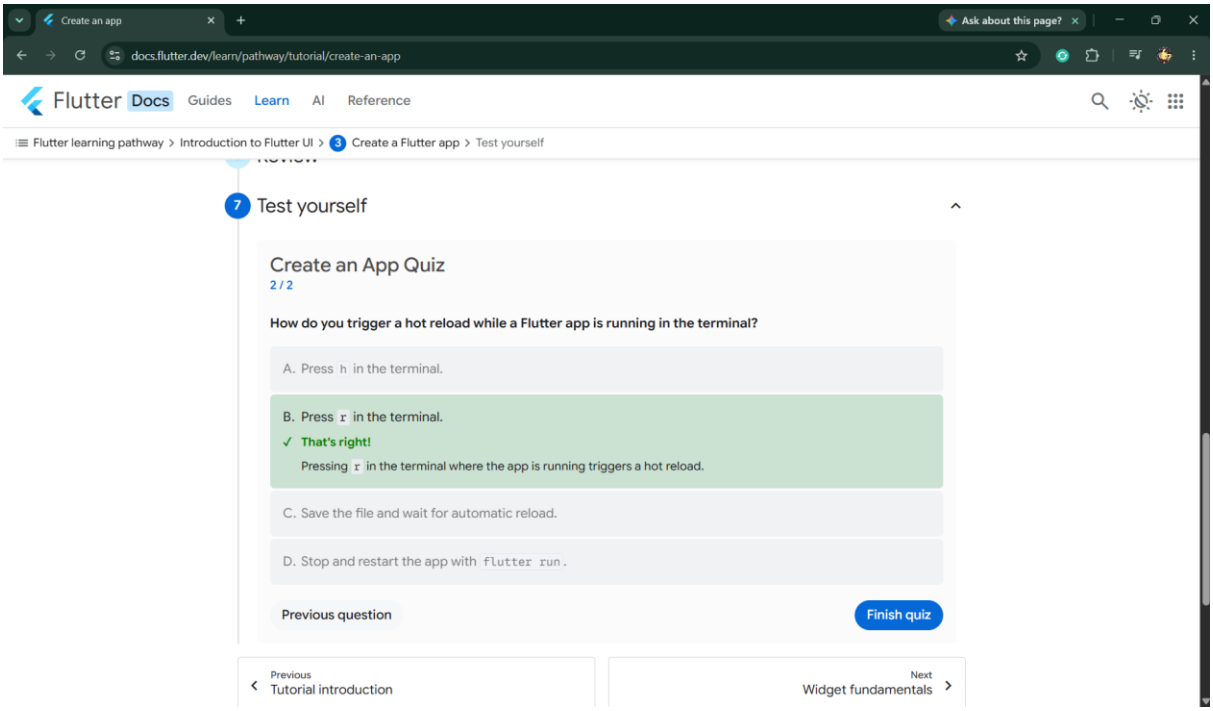


Hot reload


localhost:50356
Toi hoc flutter!!!!!!!!!!!!
main.dart
class MainApp extends StatelessWidget {
const MainApp({super.key});
@override
Widget build(BuildContext context) {
return const MaterialApp(
home: Scaffold(body: Center(child: Text('Toi hoc flutter!!!!!!!!!!!!'))),
);
}
}
Problems Output Debug Console Ports Terminal
flutter run -d chrome
Flutter run key commands.
r Hot reload.
R Hot restart.
h List all available interactive commands.
d Detach (terminate "flutter run" but leave application running).
c Clear the screen
q Quit (terminate the application on the device).
This app is linked to the debug service: ws://127.0.0.1:62519/rI5B5rnMw3s=/ws
Debug service listening on ws://127.0.0.1:62519/rI5B5rnMw3s=/ws
A Dart VM Service on Chrome is available at: http://127.0.0.1:62519/rI5B5rnMw3s=
The Flutter DevTools debugger and profiler on Chrome is available at:
http://127.0.0.1:62519/rI5B5rnMw3s=/devtools/?uri=ws://127.0.0.1:62519/rI5B5rnMw3s=/ws
Starting application from main method in: org-dartlang-app:///web_entrypoint.dart.
Performing hot reload... 605ms
Reloaded application in 607ms.

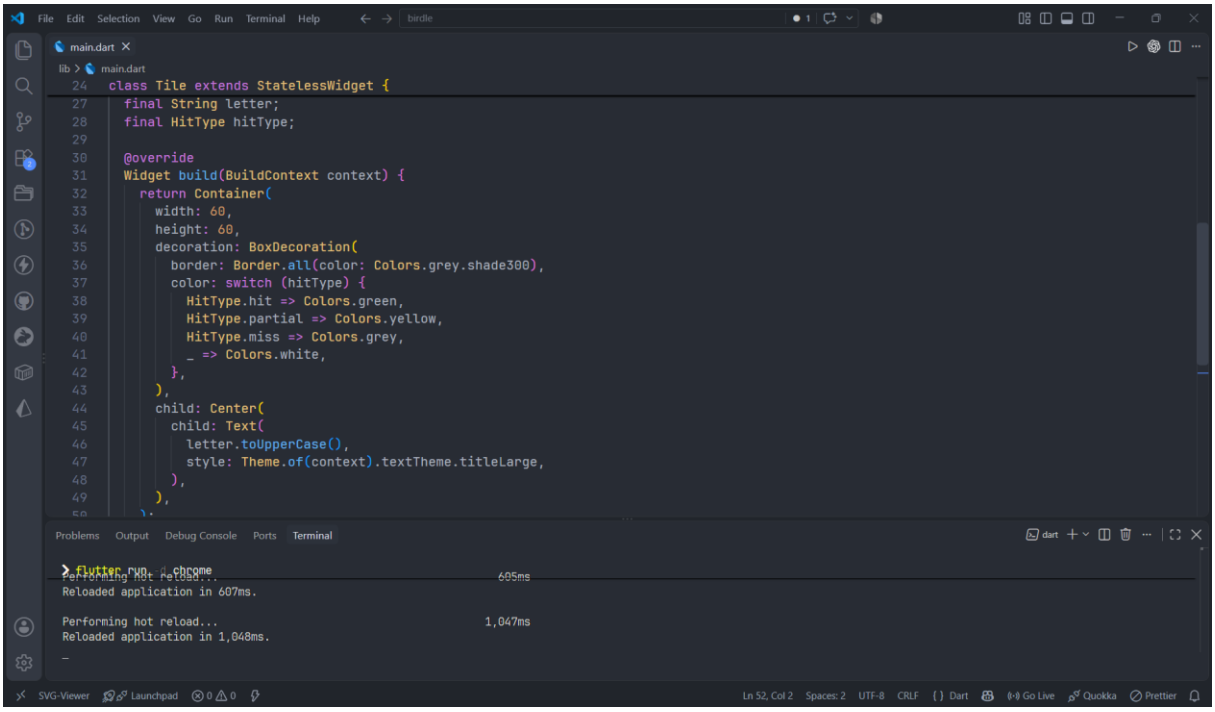
Quiz


Create an app
docs.flutter.dev/learn/pathway/tutorial/create-an-app
Flutter Docs Guides Learn AI Reference
Flutter learning pathway > Introduction to Flutter UI > 3 Create a Flutter app > Test yourself
7 Test yourself
Create an App Quiz
1 / 2
What is the purpose of the runApp function in a Flutter application?
A. It compiles the Dart code into native machine code.
B. It downloads Flutter dependencies from the internet.
C. It creates the main.dart file for the project.
D. It takes a widget as an argument and makes it the root of the widget tree.
That's right!
The runApp function inflates the given widget and attaches it to the screen, making it the root of the widget tree.
Previous question
Next question
Previous
Tutorial introduction
Next
Widget fundamentals



4/ Widget fundamentals

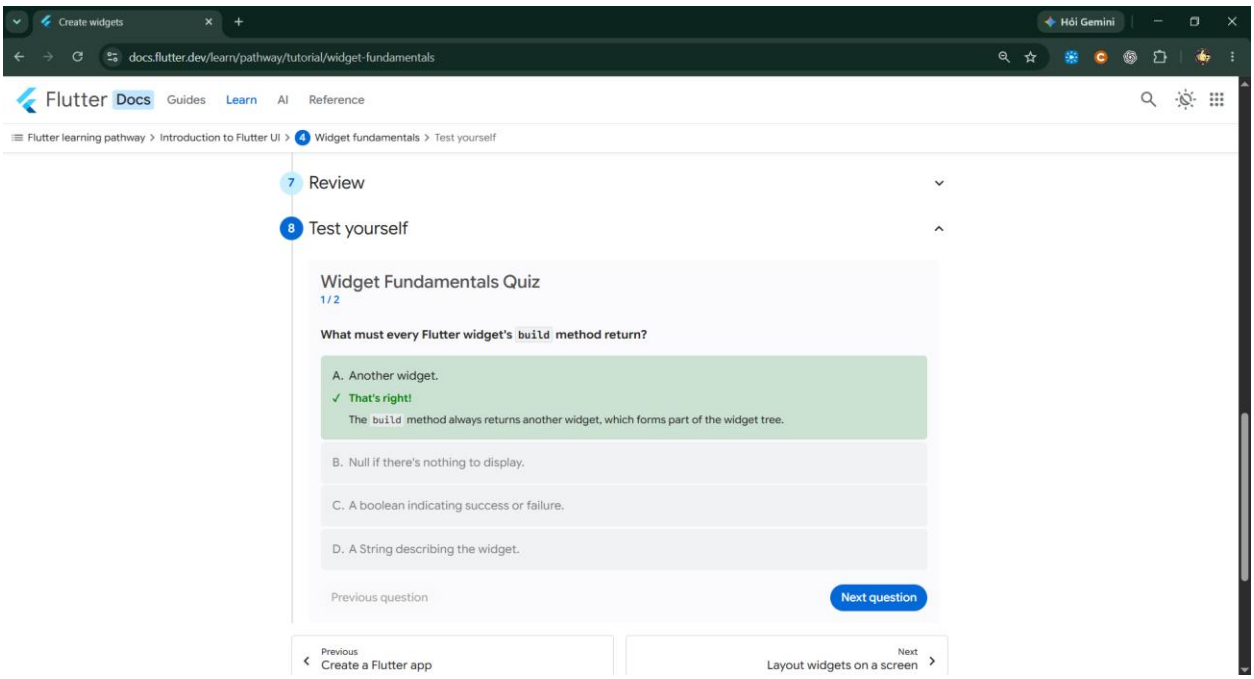
Code

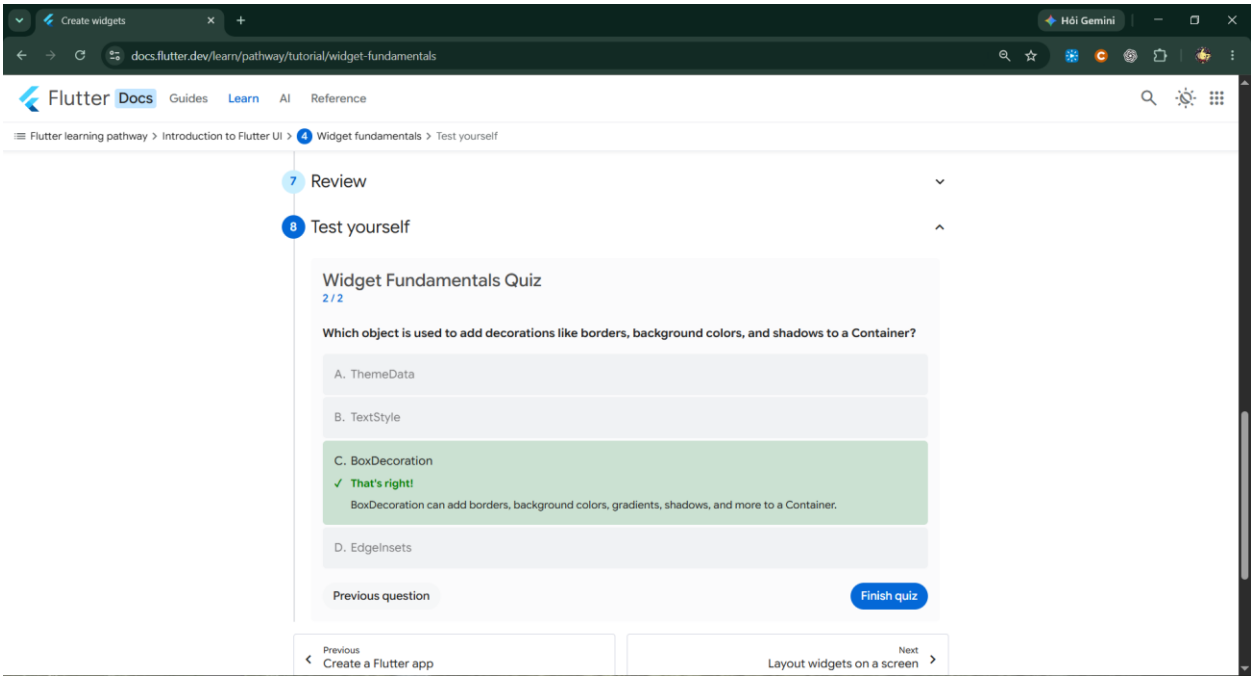


Output


localhost:50356
localhost:50356
DEBUG
A

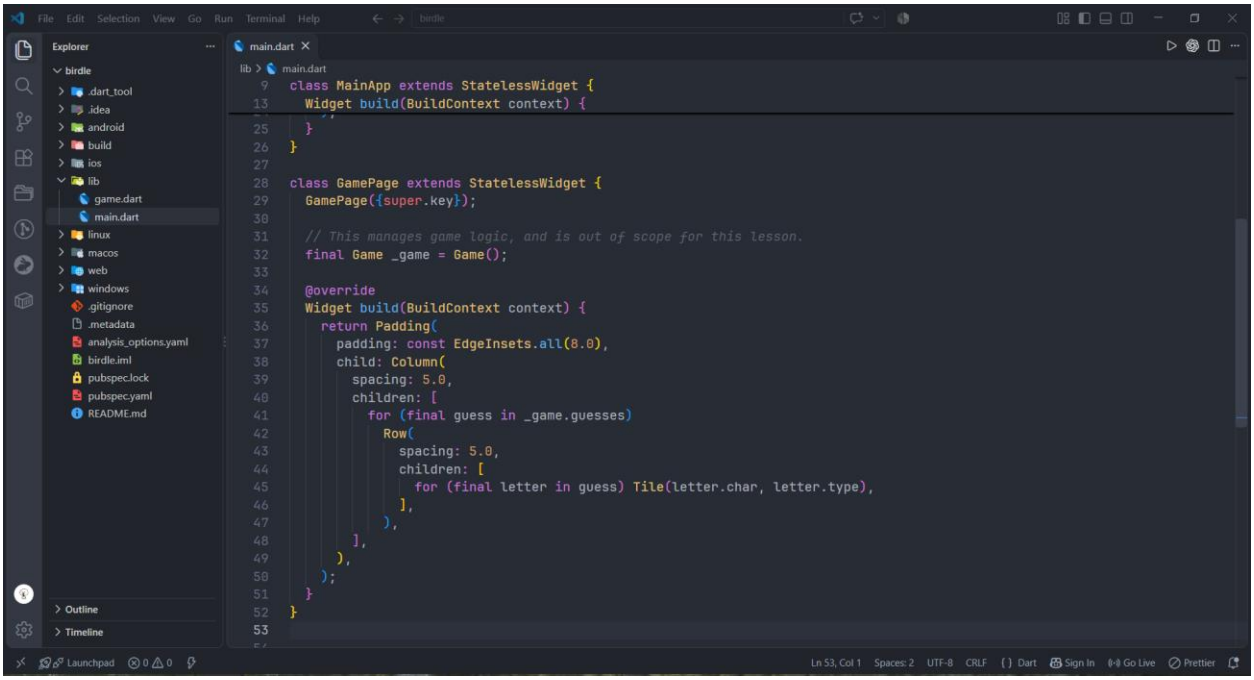
Quiz


Create widgets
Hỏi Gemini
docs.flutter.dev/learn/pathway/tutorial/widget-fundamentals
Flutter Docs Guides Learn AI Reference
Flutter learning pathway > Introduction to Flutter UI > 4 Widget fundamentals > Test yourself
7 Review
8 Test yourself
Widget Fundamentals Quiz
1 / 2
What must every Flutter widget's build method return?
A. Another widget.
✓ That's right!
The build method always returns another widget, which forms part of the widget tree.
B. Null if there's nothing to display.
C. A boolean indicating success or failure.
D. A String describing the widget.
Previous question
Next question
Previous
Create a Flutter app
Next
Layout widgets on a screen

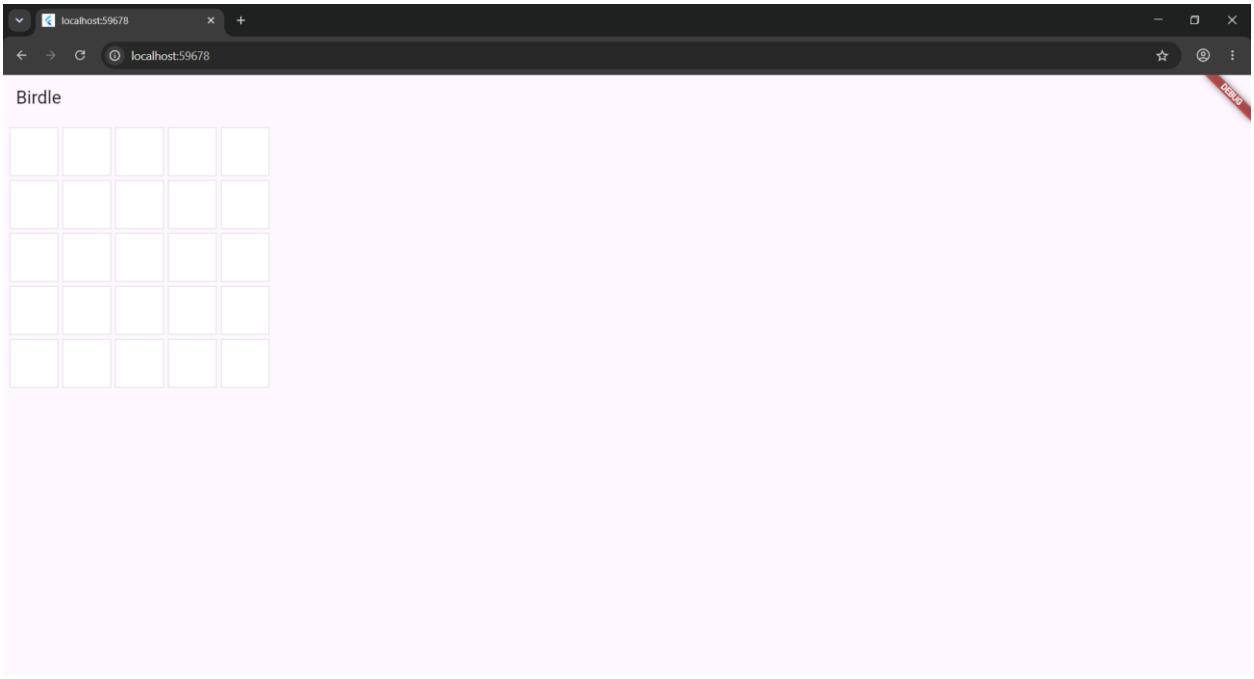


5/ Layout widgets on a screen

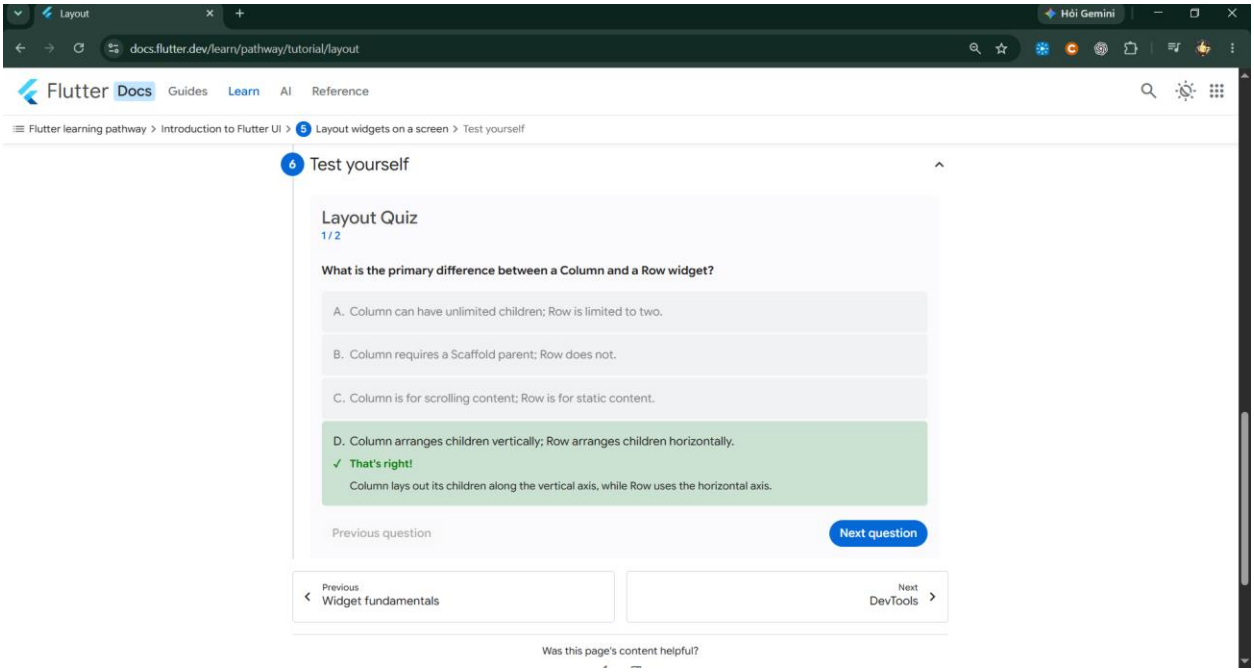
Code

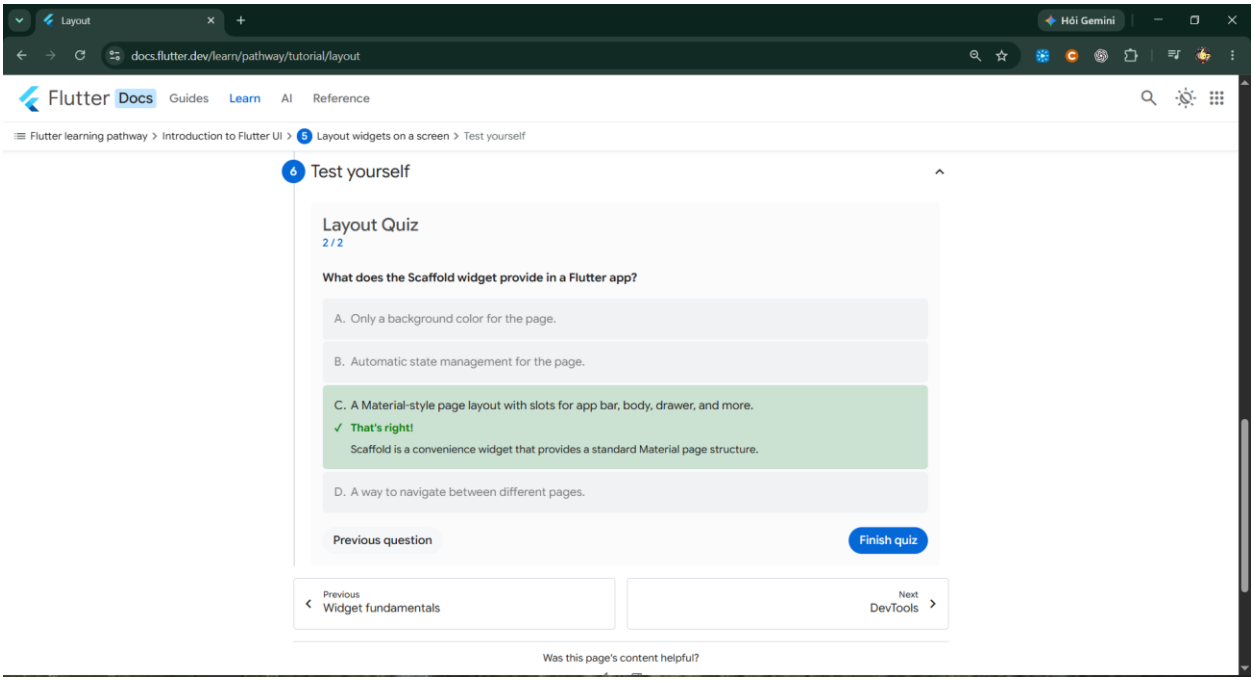


Output


localhost:59678
localhost:59678
Birdle
DEBUG

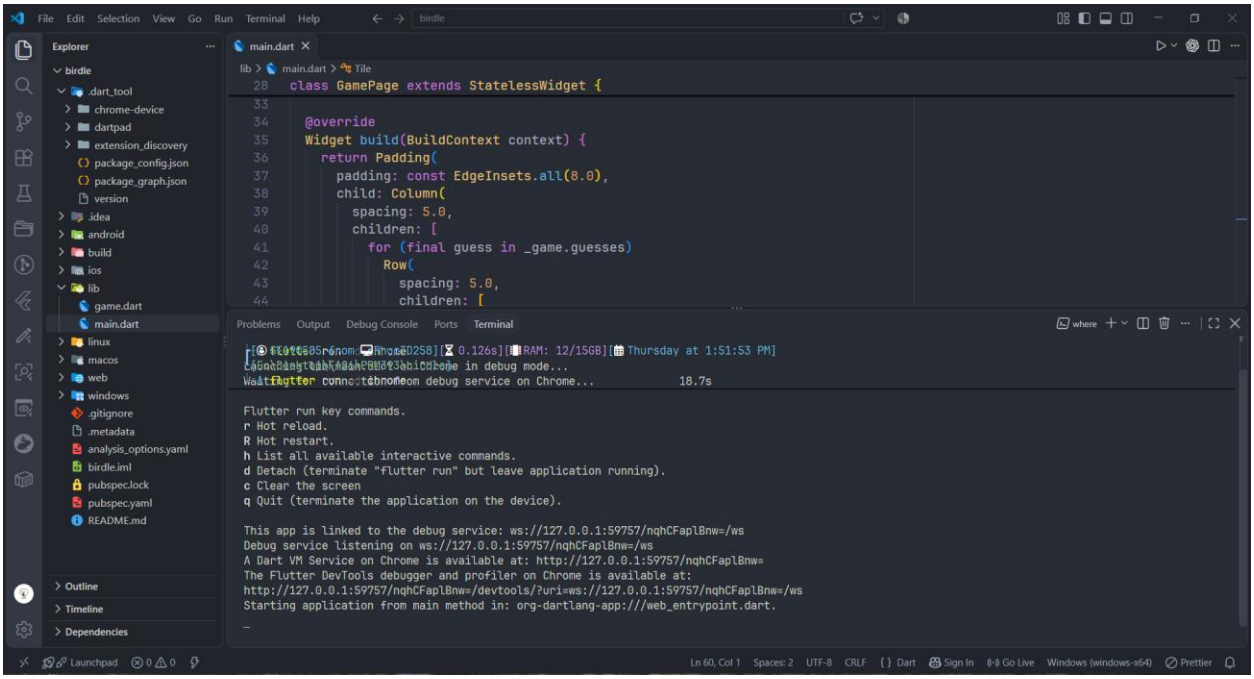
Quiz

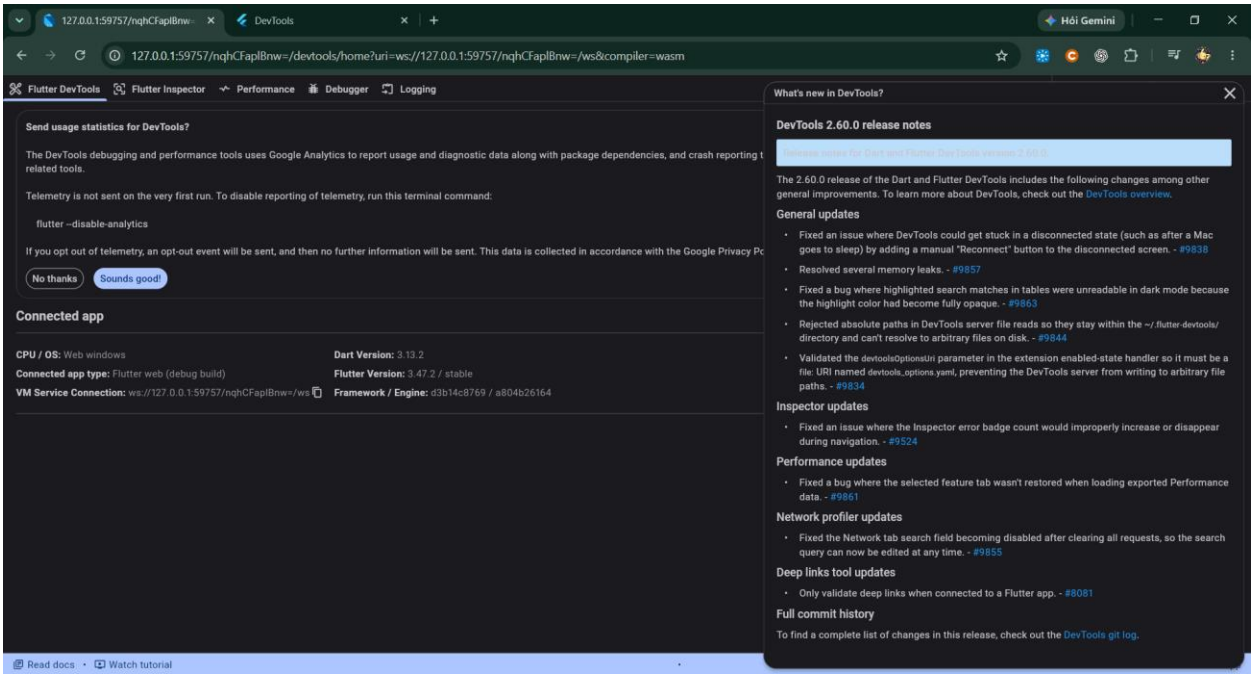

Layout
Hỏi Gemini
docs.flutter.dev/learn/pathway/tutorial/layout
Flutter Docs
Guides
Learn
AI
Reference
Flutter learning pathway > Introduction to Flutter UI > 5 Layout widgets on a screen > Test yourself
6 Test yourself
Layout Quiz
1 / 2
What is the primary difference between a Column and a Row widget?
A. Column can have unlimited children; Row is limited to two.
B. Column requires a Scaffold parent; Row does not.
C. Column is for scrolling content; Row is for static content.
D. Column arranges children vertically; Row arranges children horizontally.
That's right!
Column lays out its children along the vertical axis, while Row uses the horizontal axis.
Previous question
Next question
Previous
Widget fundamentals
Next
DevTools
Was this page's content helpful?



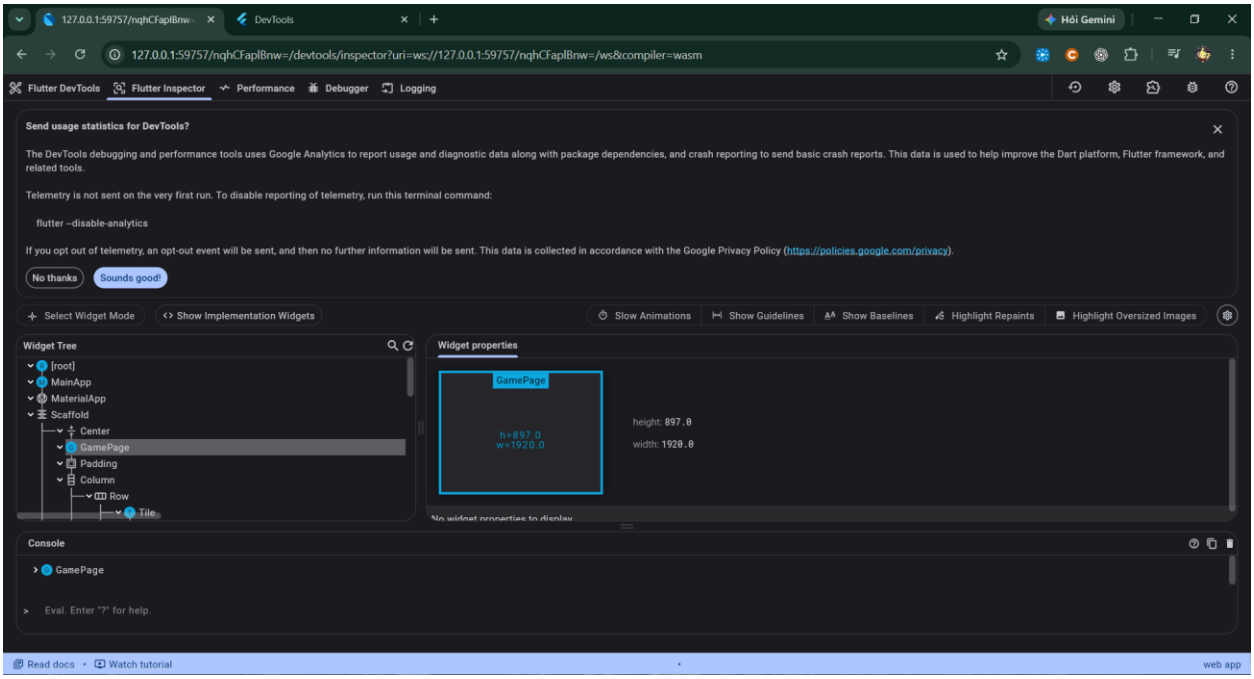
6/ DevTools

Terminal

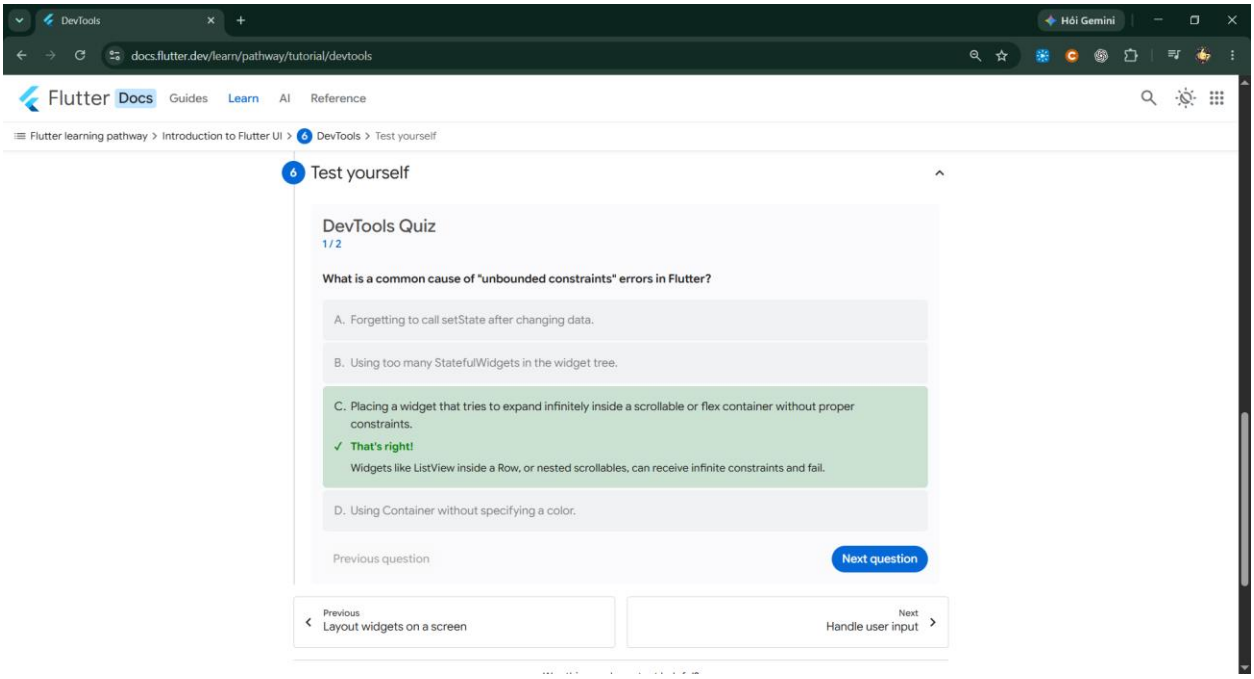


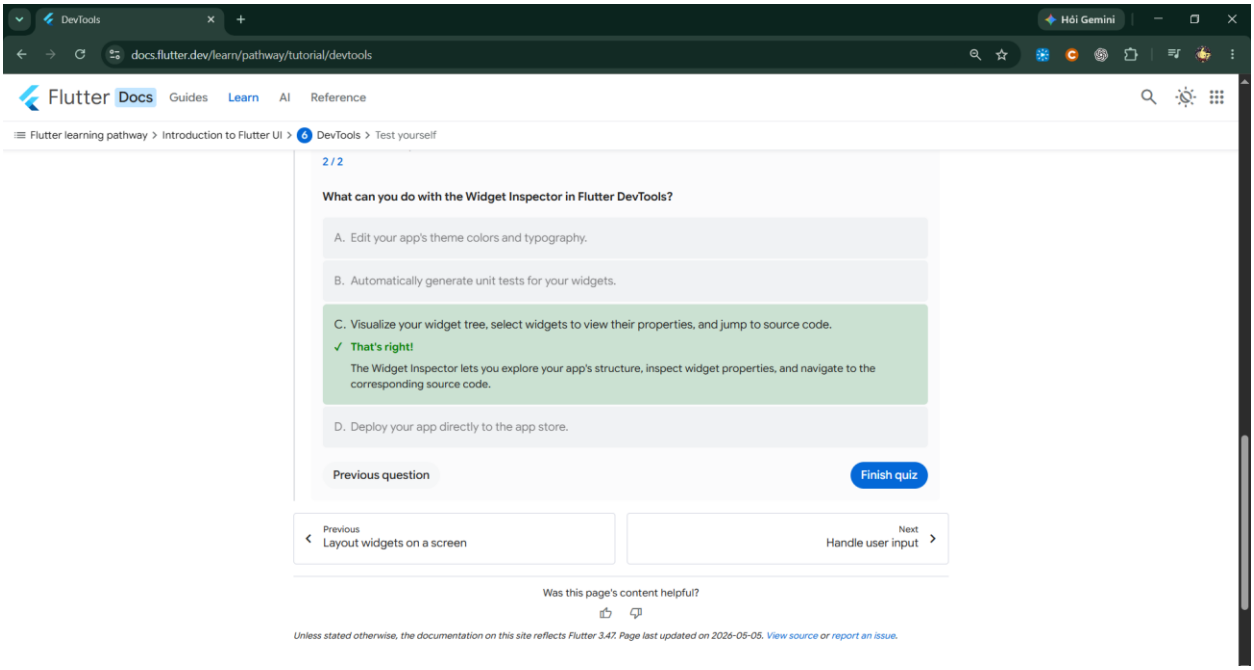


Devtools



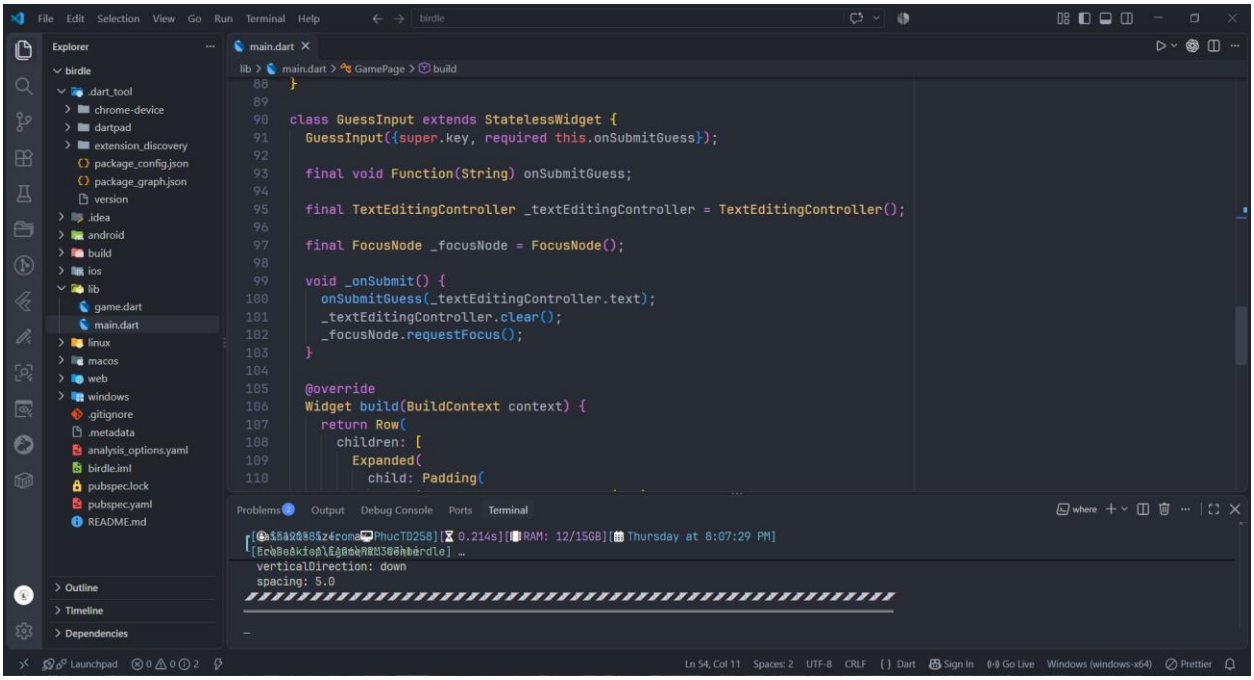
Quiz





7/ Handle user input

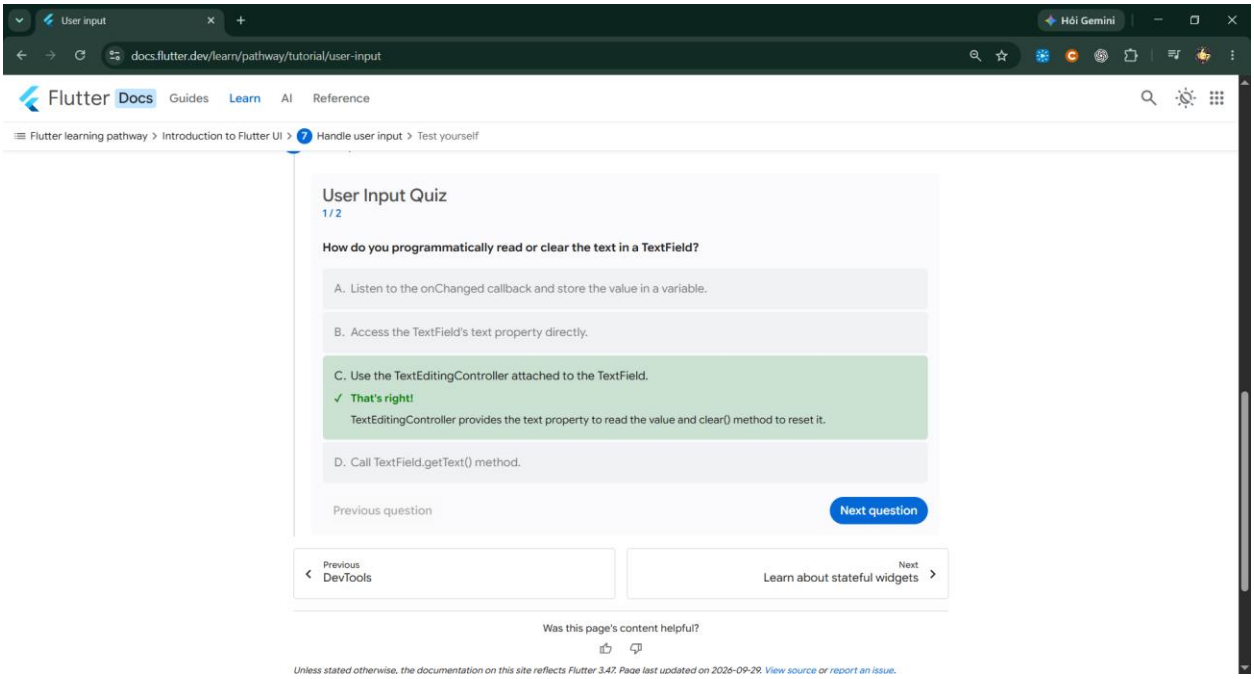
Code

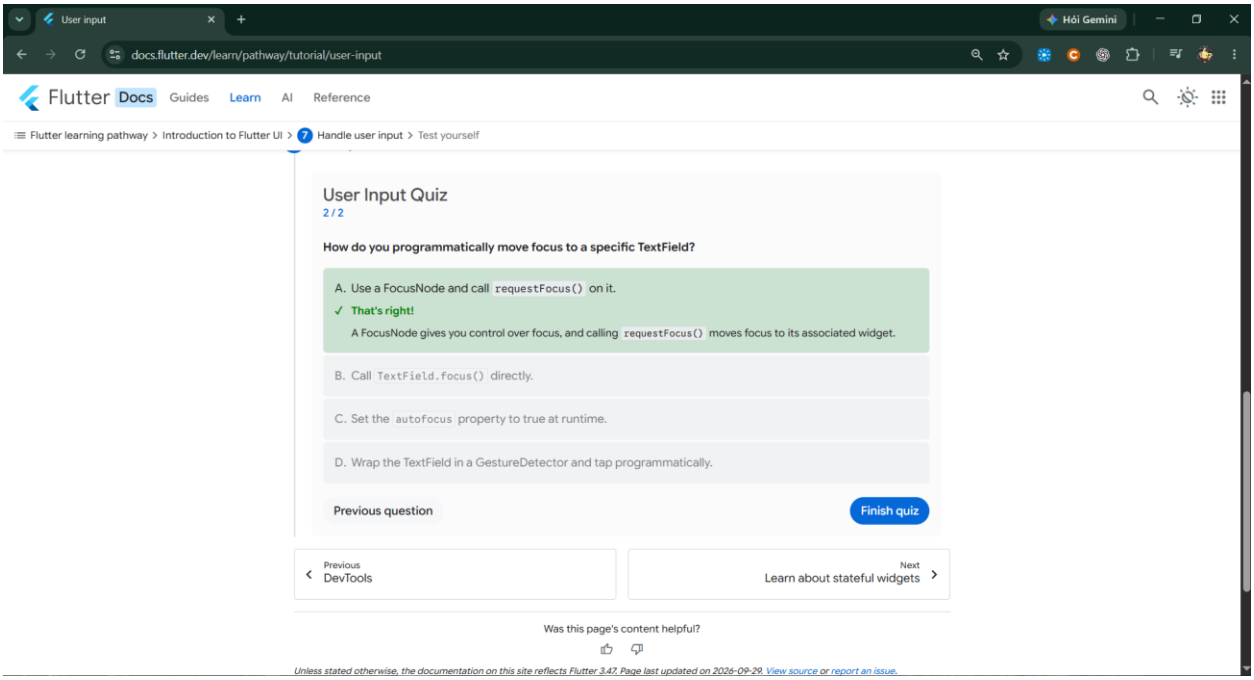

File Edit Selection View Go Run Terminal Help
birdle
Explorer
birdle
.dart_tool
chrome-device
dartpad
extension_discovery
package_config.json
package_graph.json
version
.idea
android
build
ios
lib
game.dart
main.dart
linux
macos
web
windows
.gitignore
.metadata
analysis_options.yaml
birdle.iml
pubspec.lock
pubspec.yaml
README.md
Outline
Timeline
Dependencies
main.dart
lib > main.dart > GamePage > build
88 }
89
90 class GuessInput extends StatelessWidget {
91 GuessInput({super.key, required this.onSubmitGuess});
92
93 final void Function(String) onSubmitGuess;
94
95 final TextEditingController _textEditingController = TextEditingController();
96
97 final FocusNode _focusNode = FocusNode();
98
99 void _onSubmit() {
100 onSubmitGuess(_textEditingController.text);
101 _textEditingController.clear();
102 _focusNode.requestFocus();
103 }
104
105 @override
106 Widget build(BuildContext context) {
107 return Row(
108 children: [
109 Expanded(
110 child: Padding(
Problems
Output
Debug Console
Ports
Terminal
where
Thursday at 8:07:29 PM]
verticalDirection: down
spacing: 5.0
Launchpad
Ln 54, Col 11
Spaces: 2
UTF-8
CRLF
Dart
Sign In
Go Live
Windows (windows-x64)
Prettier

Output


localhost:61744
localhost:61744
Birdle
DEBUG
0/5

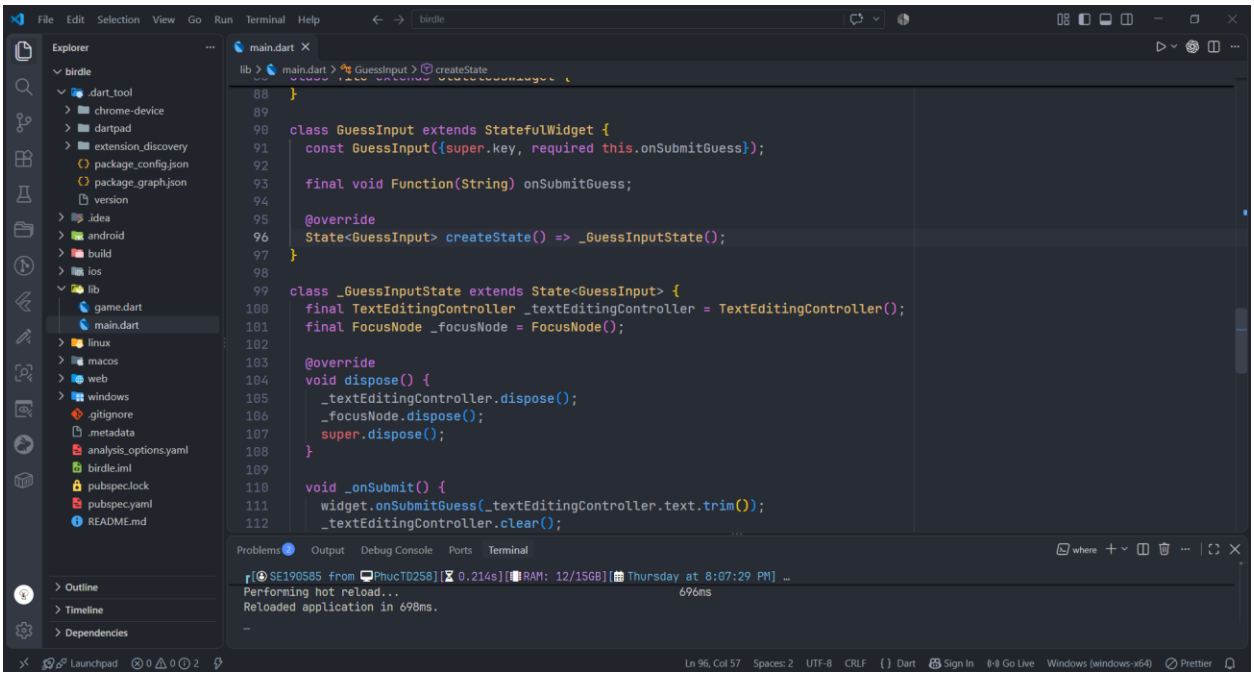
Quiz



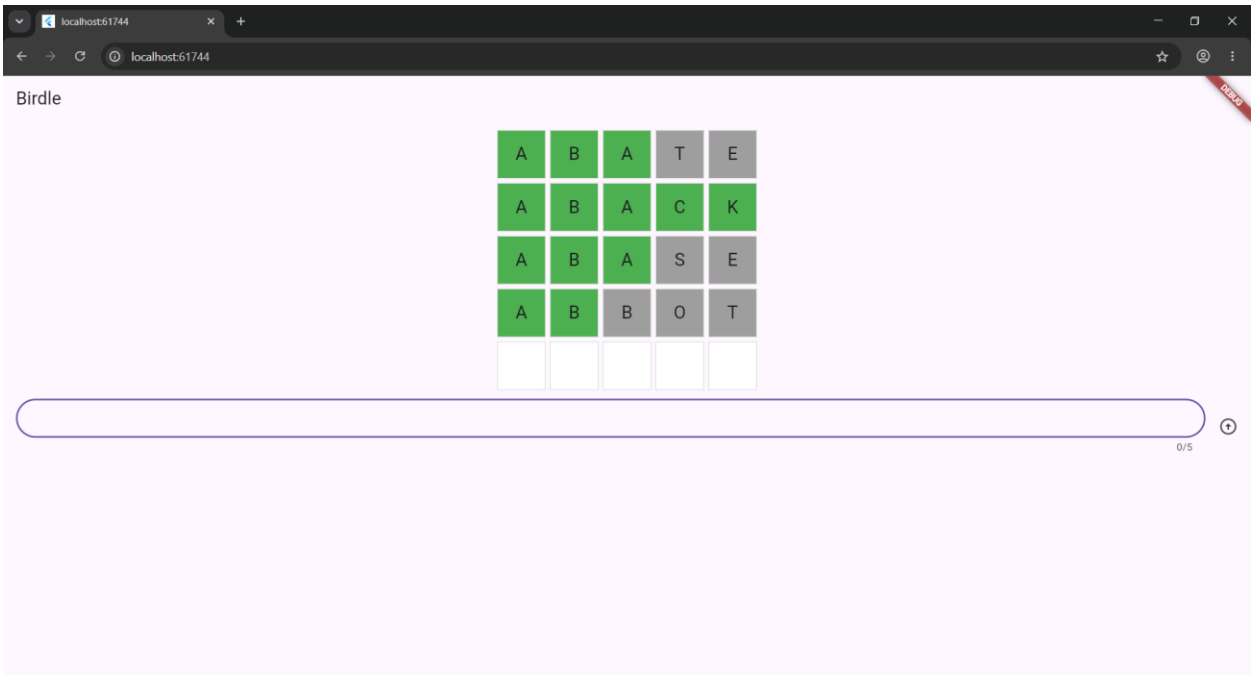


8/ Learn about stateful widgets

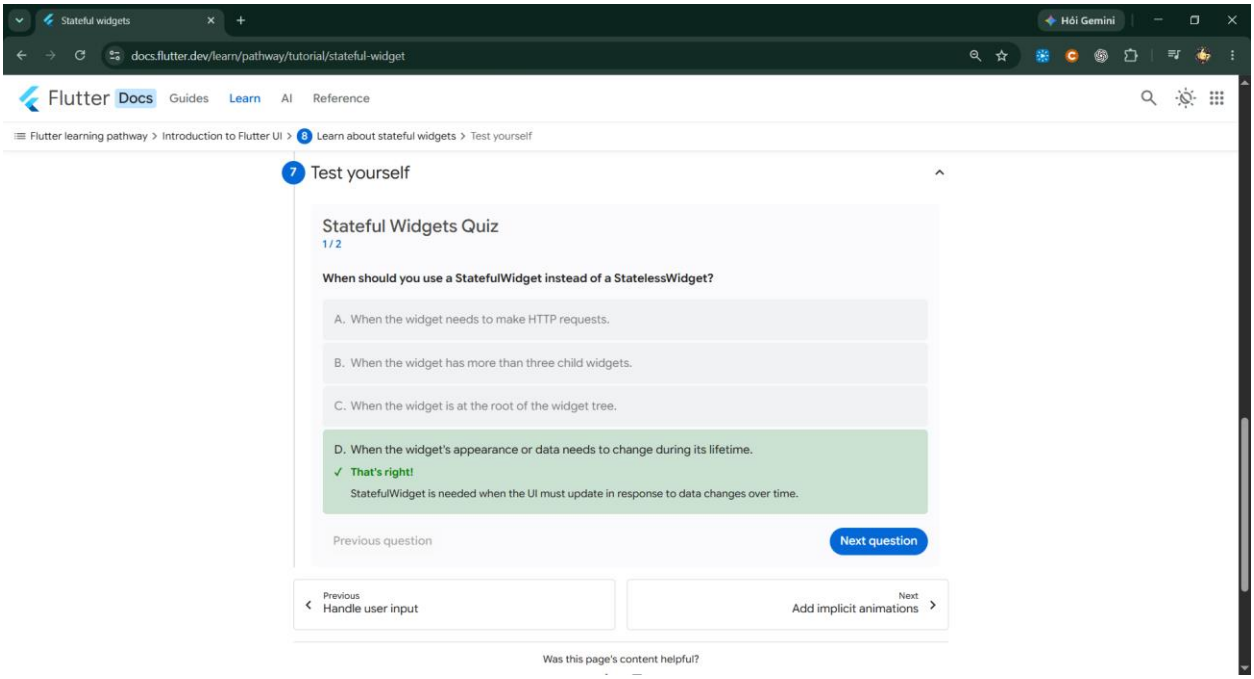
Code

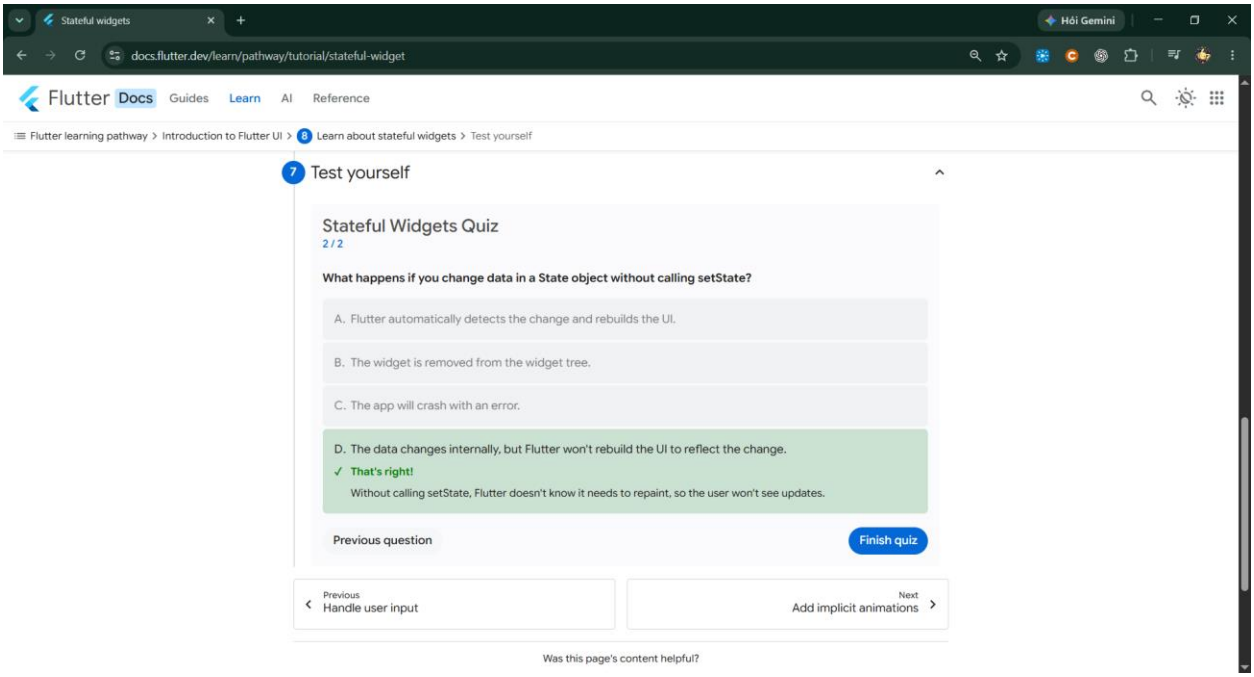

File Edit Selection View Go Run Terminal Help
birdle
Explorer
main.dart
lib > main.dart > GuessInput > createState
88 }
89
90 class GuessInput extends StatefulWidget {
91 const GuessInput({super.key, required this.onSubmitGuess});
92
93 final void Function(String) onSubmitGuess;
94
95 @override
96 State<GuessInput> createState() => _GuessInputState();
97 }
98
99 class _GuessInputState extends State<GuessInput> {
100 final TextEditingController _textEditingController = TextEditingController();
101 final FocusNode _focusNode = FocusNode();
102
103 @override
104 void dispose() {
105 _textEditingController.dispose();
106 _focusNode.dispose();
107 super.dispose();
108 }
109
110 void _onSubmit() {
111 widget.onSubmitGuess(_textEditingController.text.trim());
112 _textEditingController.clear();
birdle
.dart_tool
chrome-device
dartpad
extension_discovery
package_config.json
package_graph.json
version
.idea
android
build
ios
lib
game.dart
main.dart
linux
macos
web
windows
.gitignore
.metadata
analysis_options.yaml
birdle.iml
pubspec.lock
pubspec.yaml
README.md
Outline
Timeline
Dependencies
Problems
Output
Debug Console
Ports
Terminal
Performing hot reload...
696ms
Reloaded application in 698ms.
Launchpad
Ln 96, Col 57
Spaces: 2
UTF-8
CRLF
Dart
Sign In
Go Live
Windows (windows-x64)
Prettier

Output


localhost:61744
Birdle
A B A T E
A B A C K
A B A S E
A B B O T
0/5

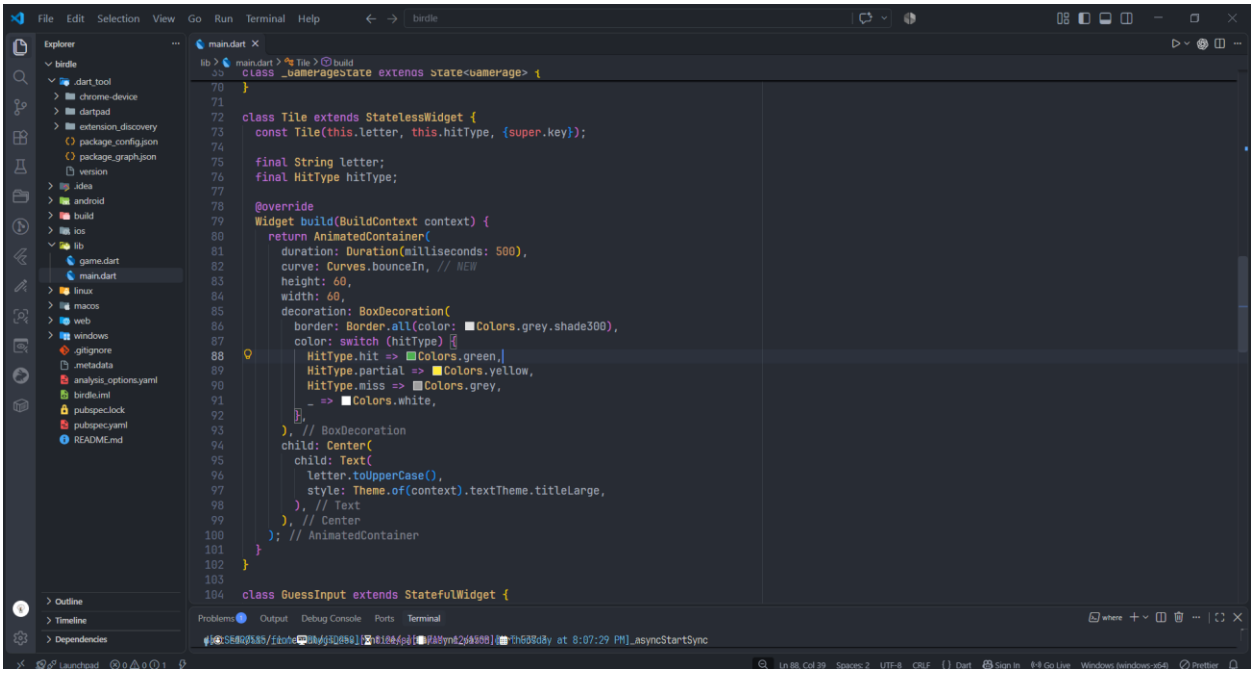
Quiz





9/ Add implicit animations

Code



```
class _GameStageState extends State<GameStage> {
}

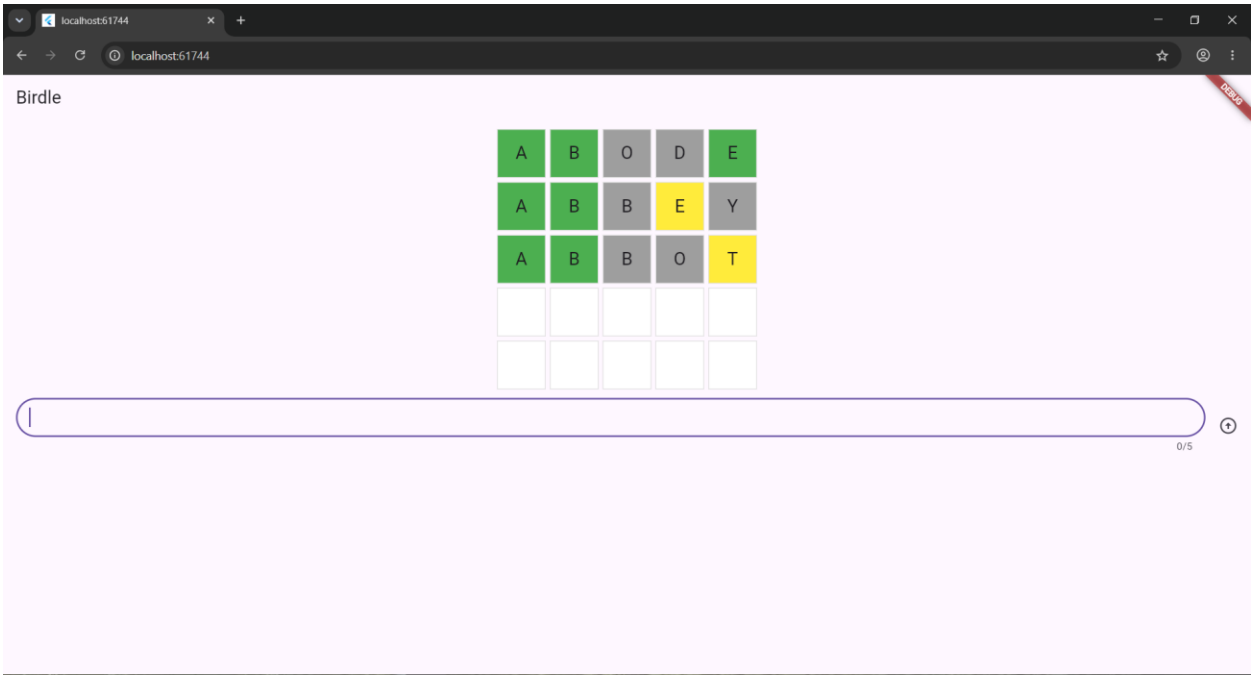
class Tile extends StatelessWidget {
  const Tile(this.letter, this.hitType, {super.key});

  final String letter;
  final HitType hitType;

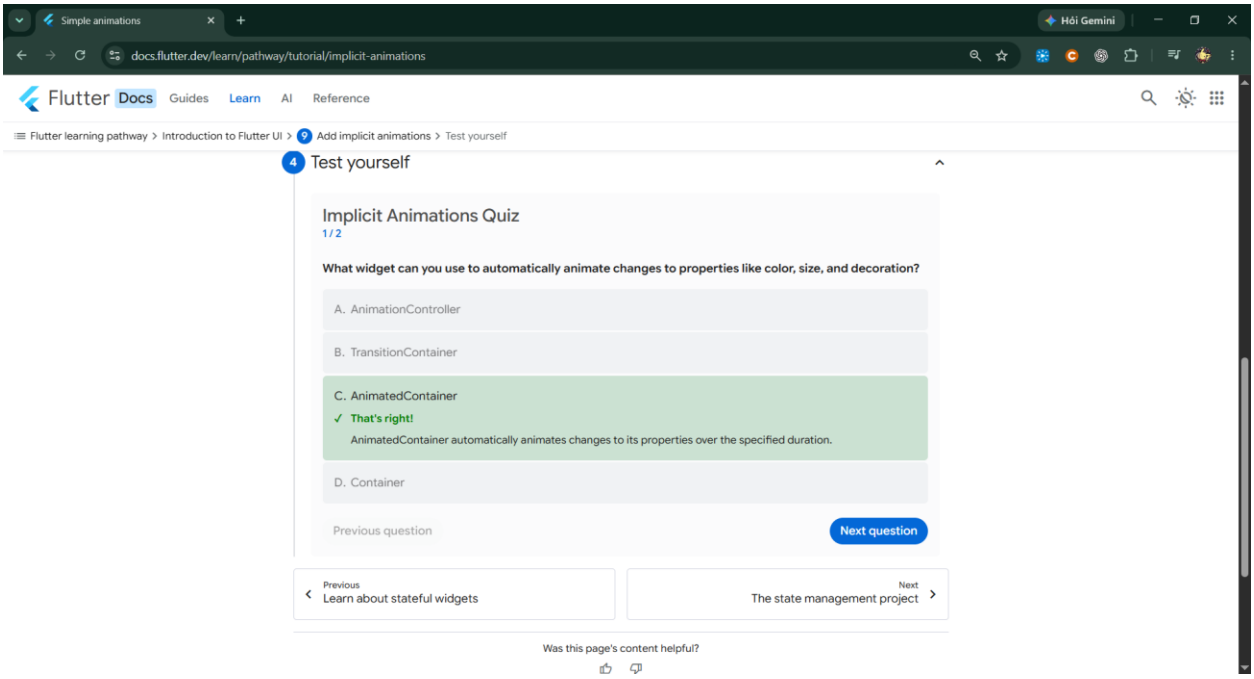
  @override
  Widget build(BuildContext context) {
    return AnimatedContainer(
      duration: Duration(milliseconds: 500),
      curve: Curves.bounceIn, // NEW
      height: 60,
      width: 60,
      decoration: BoxDecoration(
        border: Border.all(color: Colors.grey.shade300),
        color: switch (hitType) {
          HitType.hit => Colors.green,
          HitType.partial => Colors.yellow,
          HitType.miss => Colors.grey,
          _ => Colors.white,
        },
      ), // BoxDecoration
      child: Center(
        child: Text(
          letter.toUpperCase(),
          style: Theme.of(context).textTheme.titleLarge,
        ), // Text
      ), // Center
    ); // AnimatedContainer
  }
}

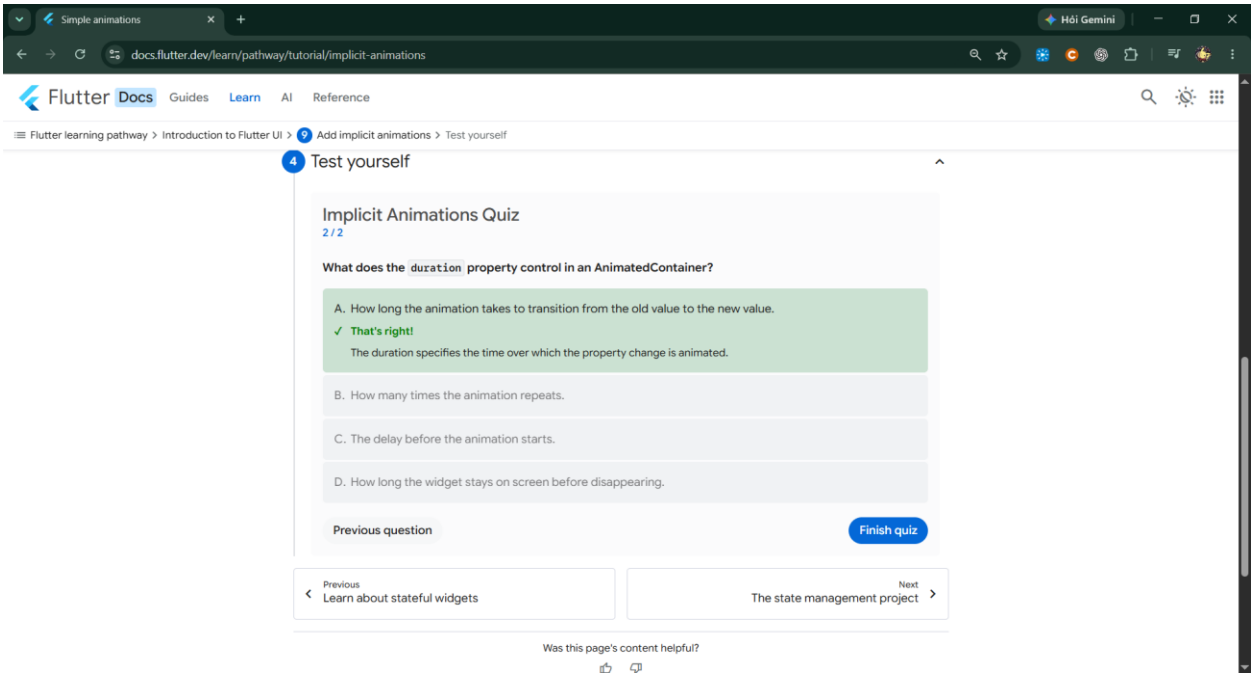
class GuessInput extends StatefulWidget {
```

Output



Quiz

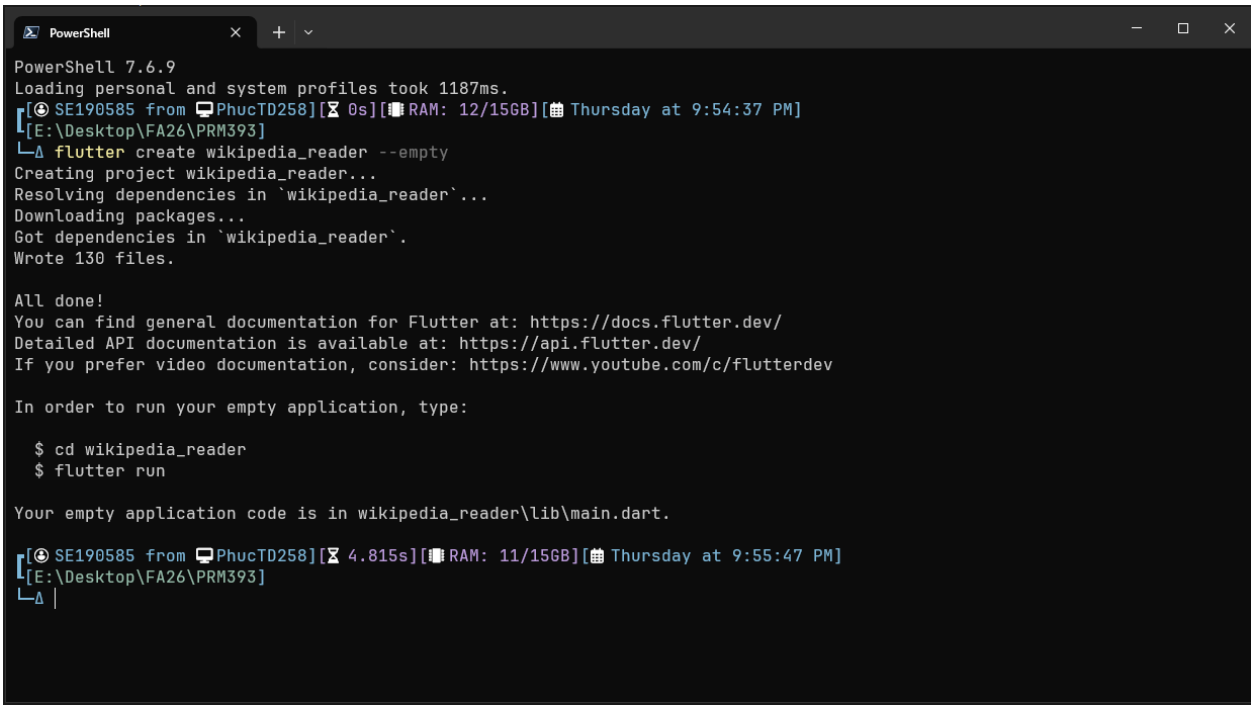




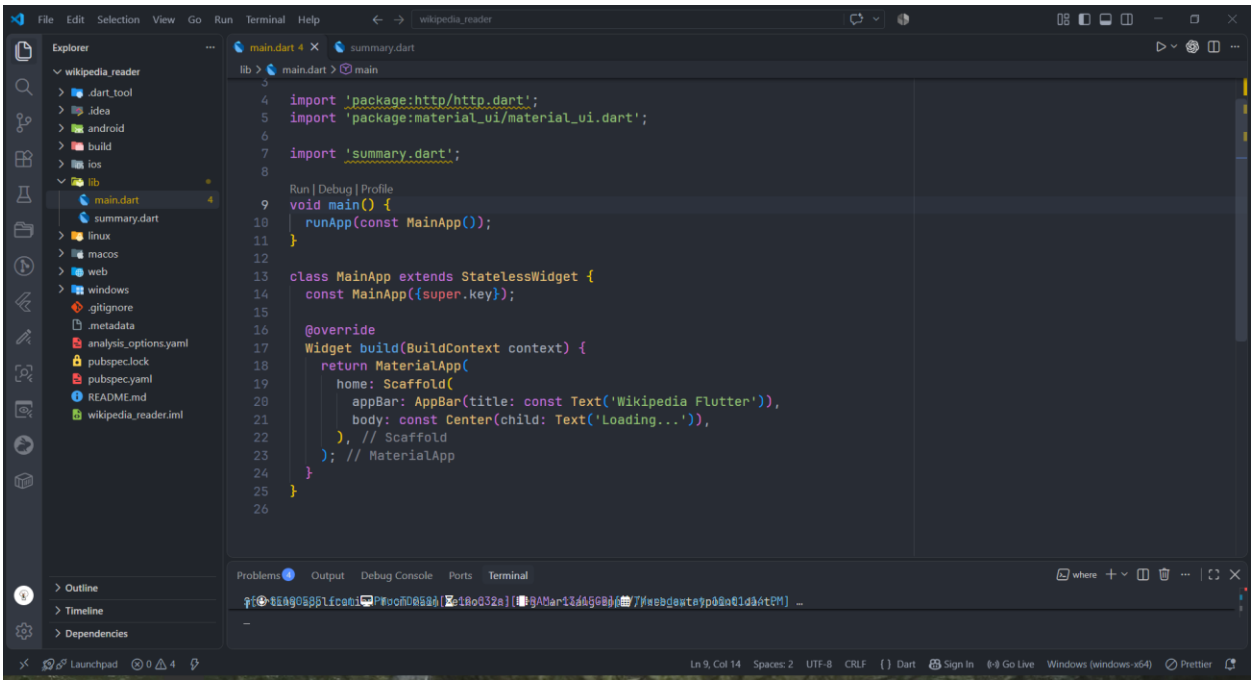
Project 2 – Wikipedia Reader (State in Flutter apps)

10/ The state management project

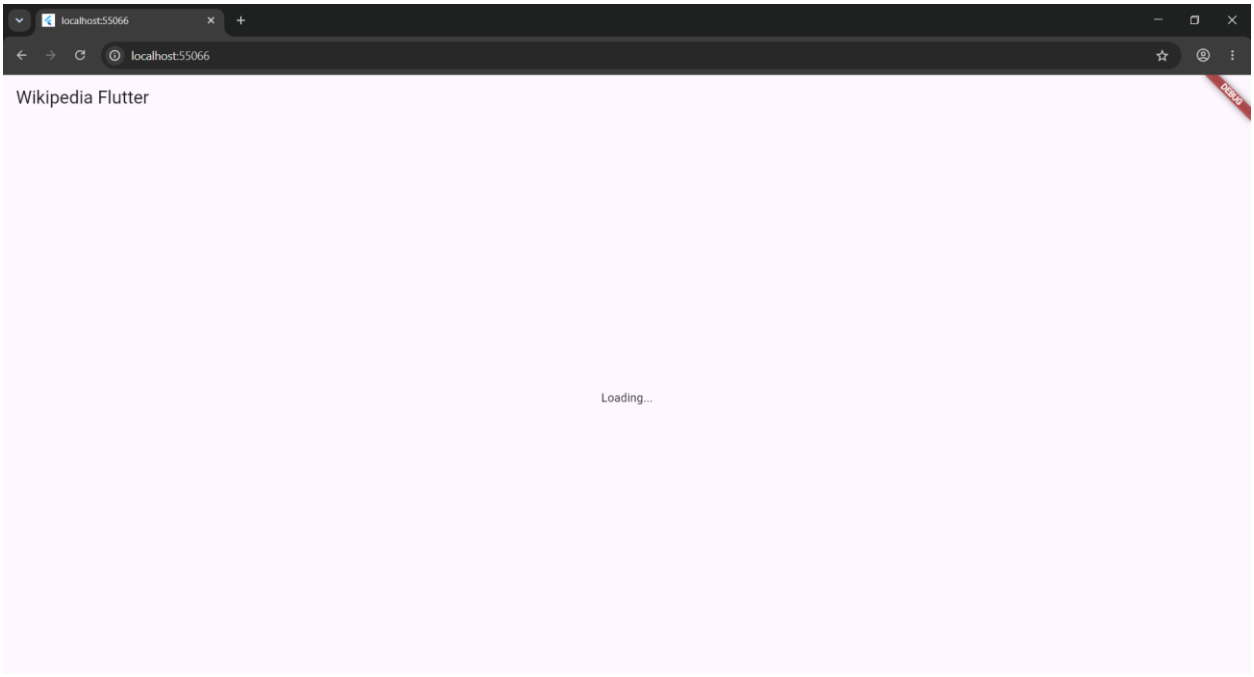
Terminal


PowerShell
PowerShell 7.6.9
Loading personal and system profiles took 1187ms.
[SE190585 from PhucTD258][0s][RAM: 12/15GB][Thursday at 9:54:37 PM]
[E:\Desktop\FA26\PRM393]
flutter create wikipedia_reader --empty
Creating project wikipedia_reader...
Resolving dependencies in `wikipedia_reader`...
Downloading packages...
Got dependencies in `wikipedia_reader`.
Wrote 130 files.
All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev
In order to run your empty application, type:
$ cd wikipedia_reader
$ flutter run
Your empty application code is in wikipedia_reader\lib\main.dart.
[SE190585 from PhucTD258][4.815s][RAM: 11/15GB][Thursday at 9:55:47 PM]
[E:\Desktop\FA26\PRM393]

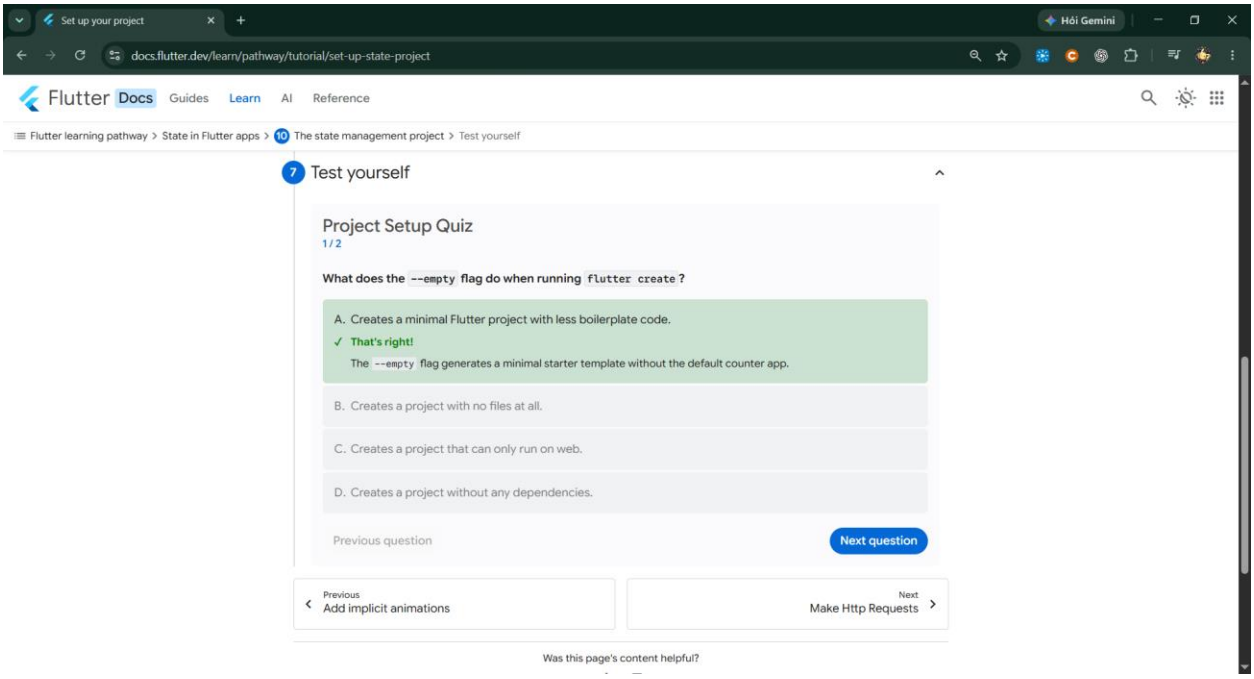
Code

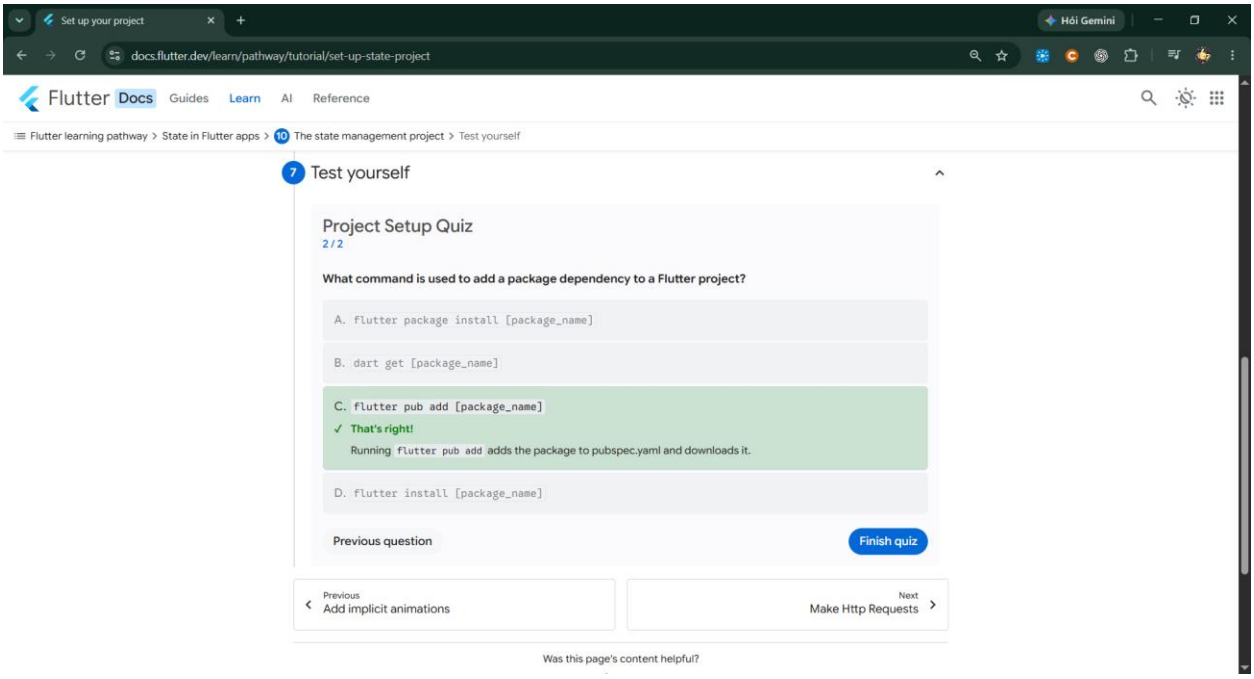

main.dart
summary.dart
import 'package:http/http.dart';
import 'package:material_ui/material_ui.dart';
import 'summary.dart';
void main() {
runApp(const MainApp());
}
class MainApp extends StatelessWidget {
const MainApp({super.key});
@override
Widget build(BuildContext context) {
return MaterialApp(
home: Scaffold(
appBar: AppBar(title: const Text('Wikipedia Flutter')),
body: const Center(child: Text('Loading...')),
), // Scaffold
); // MaterialApp
}
}

Output


localhost:55066
localhost:55066
Wikipedia Flutter
DEBUG
Loading...

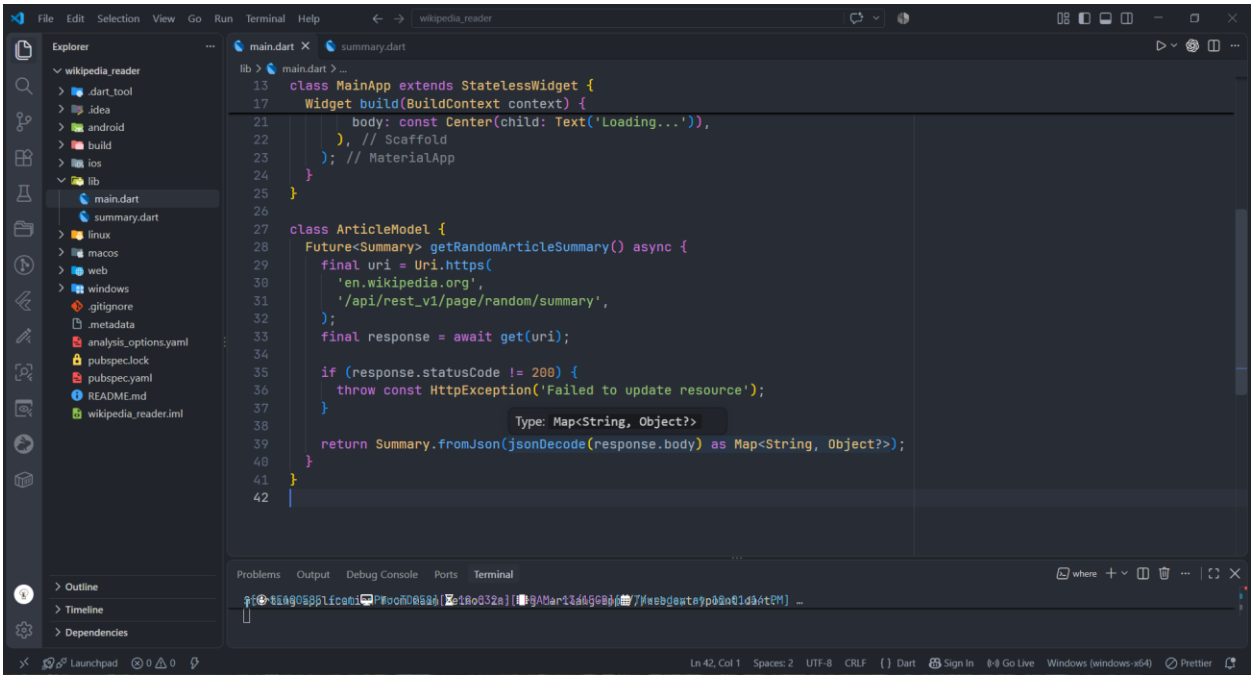
Quiz


Set up your project
Hỏi Gemini
docs.flutter.dev/learn/pathway/tutorial/set-up-state-project
Flutter Docs Guides Learn AI Reference
Flutter learning pathway > State in Flutter apps > 10 The state management project > Test yourself
7 Test yourself
Project Setup Quiz
1 / 2
What does the --empty flag do when running flutter create?
A. Creates a minimal Flutter project with less boilerplate code.
That's right!
The --empty flag generates a minimal starter template without the default counter app.
B. Creates a project with no files at all.
C. Creates a project that can only run on web.
D. Creates a project without any dependencies.
Previous question
Next question
Previous
Add implicit animations
Next
Make Http Requests
Was this page's content helpful?

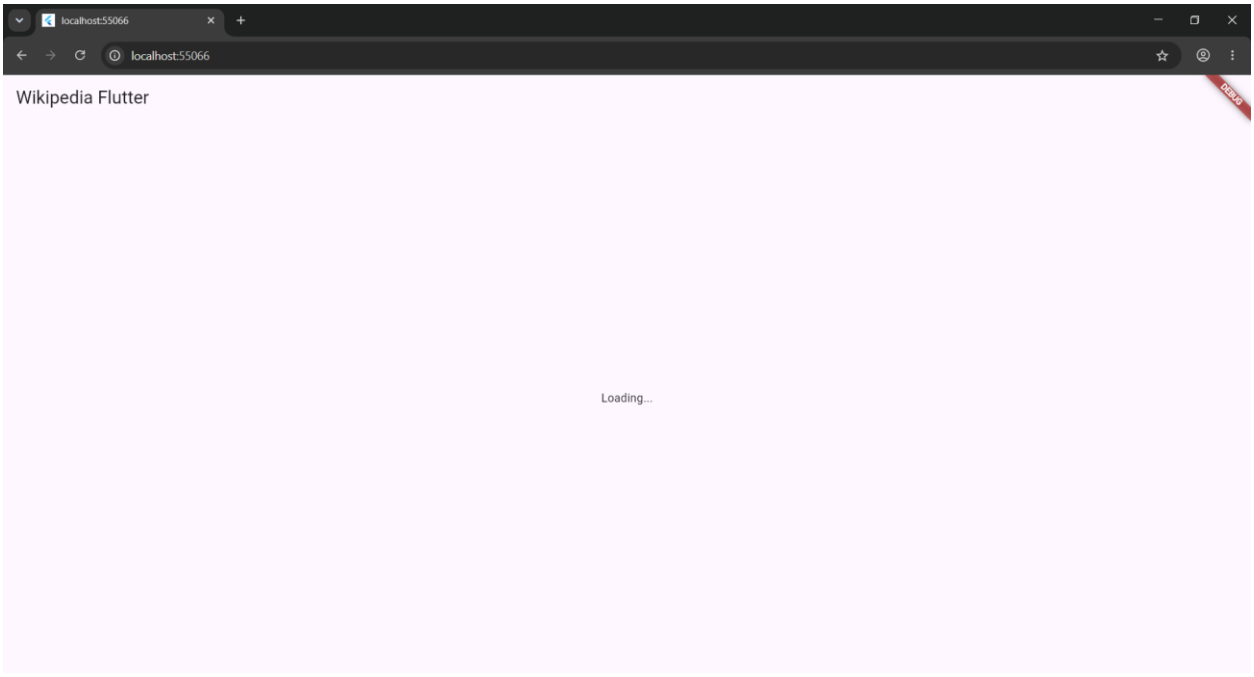


11/ Make HTTP requests

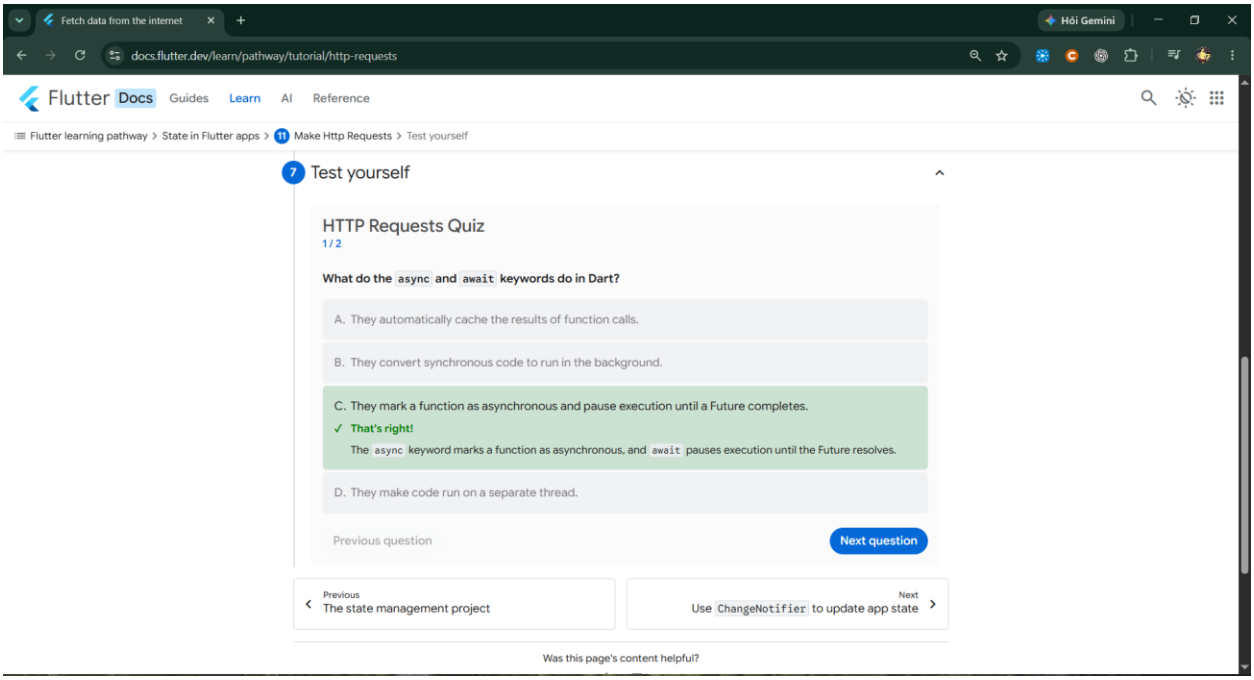
Code

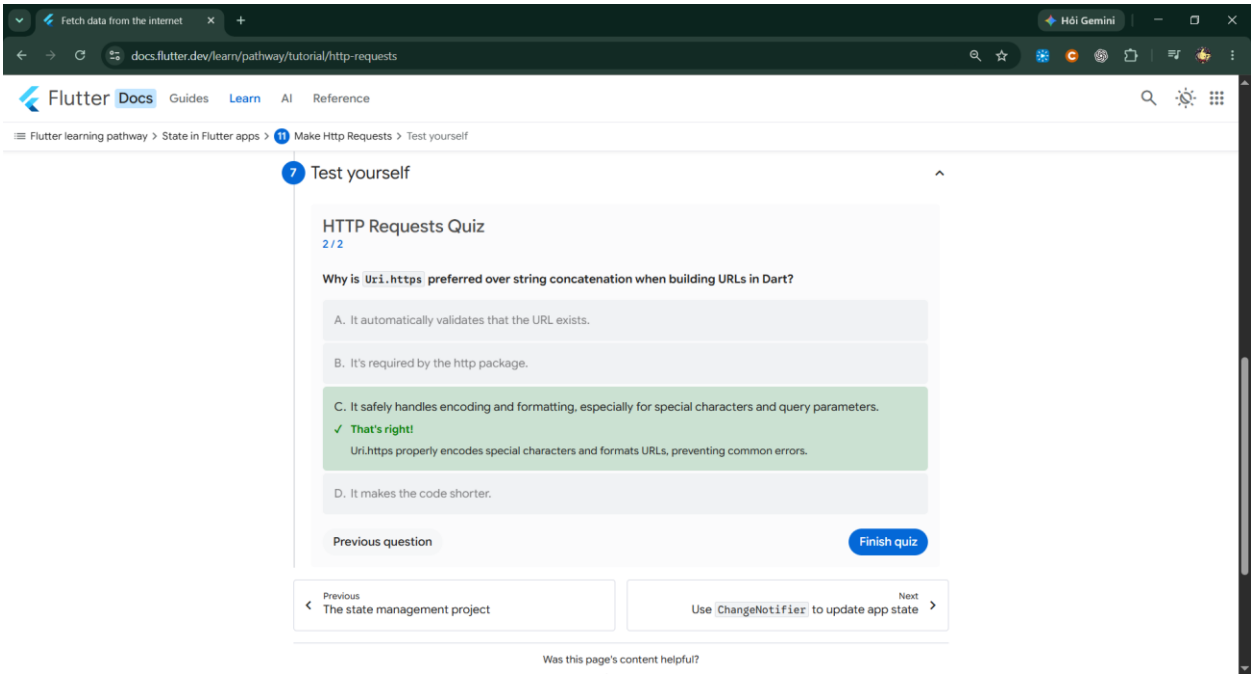


Output


localhost:55066
localhost:55066
Wikipedia Flutter
DEBUG
Loading...

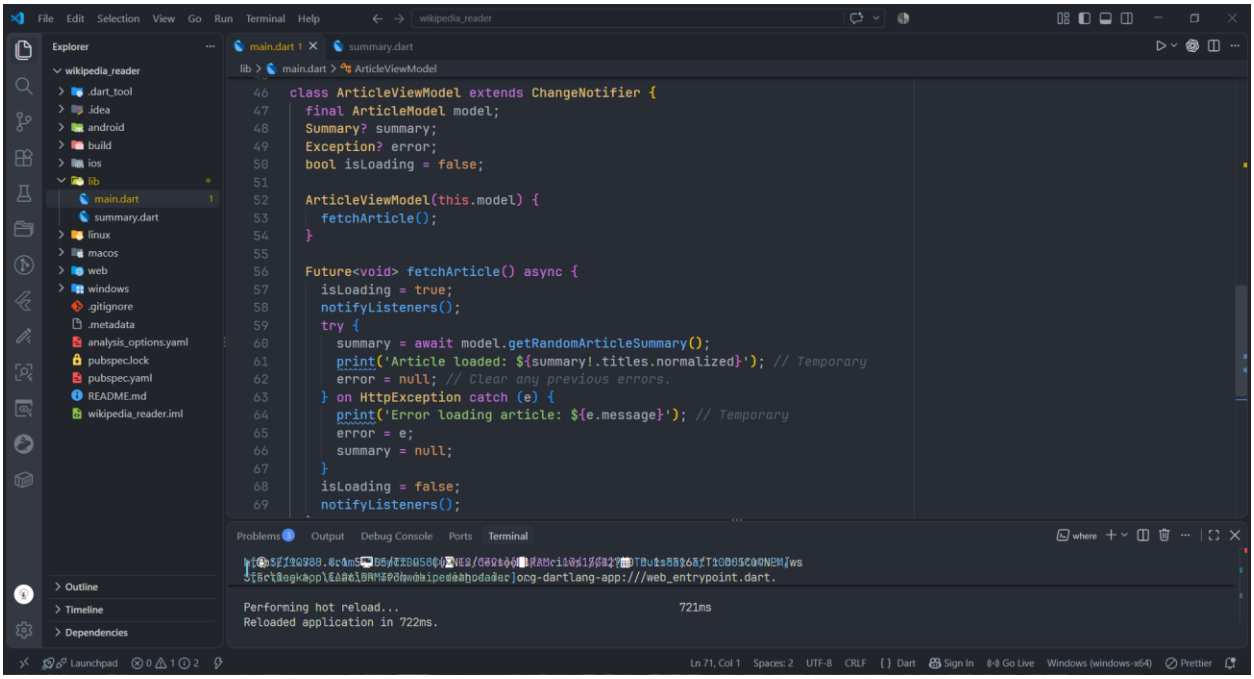
Quiz


Fetch data from the internet
Hỏi Gemini
docs.flutter.dev/learn/pathway/tutorial/http-requests
Flutter Docs Guides Learn AI Reference
Flutter learning pathway > State in Flutter apps > 11 Make Http Requests > Test yourself
7 Test yourself
HTTP Requests Quiz
1 / 2
What do the async and await keywords do in Dart?
A. They automatically cache the results of function calls.
B. They convert synchronous code to run in the background.
C. They mark a function as asynchronous and pause execution until a Future completes.
That's right!
The async keyword marks a function as asynchronous, and await pauses execution until the Future resolves.
D. They make code run on a separate thread.
Previous question
Next question
Previous
The state management project
Next
Use ChangeNotifier to update app state
Was this page's content helpful?



12/ Use ChangeNotifier to update app state

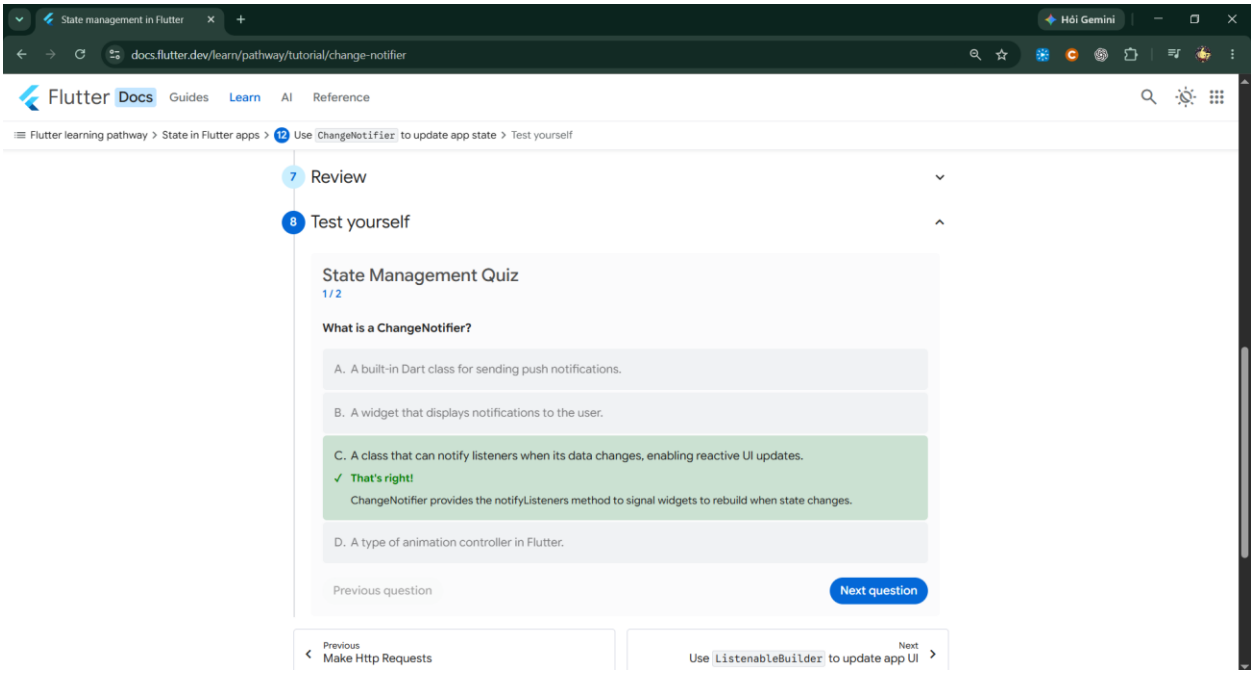
Code

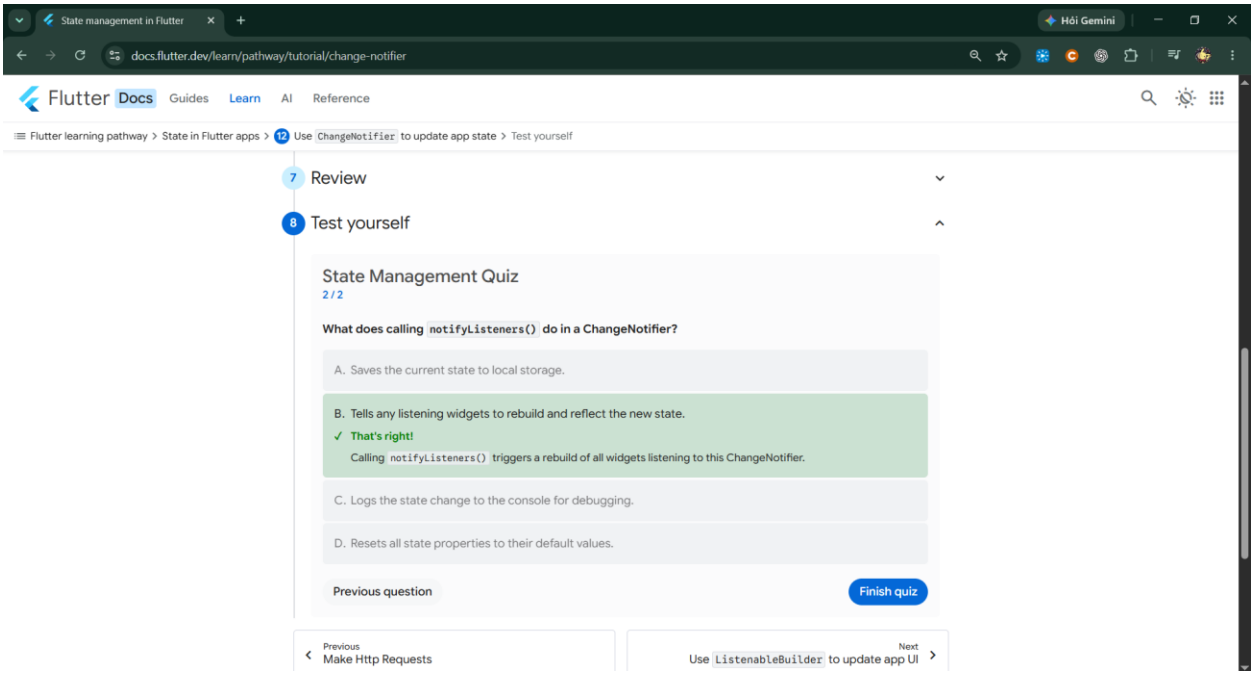


Output


localhost:55066
localhost:55066
Wikipedia Flutter
DEBUG
Check console for article data

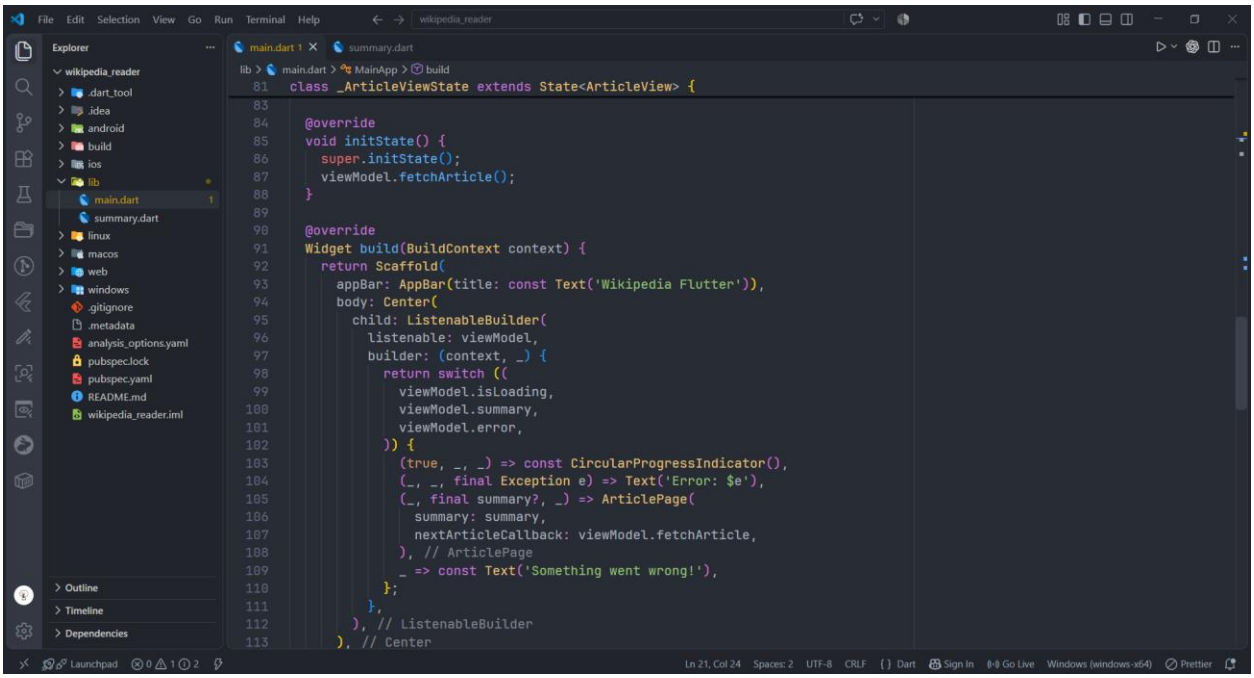
Quiz


State management in Flutter
Hỏi Gemini
docs.flutter.dev/learn/pathway/tutorial/change-notifier
Flutter Docs Guides Learn AI Reference
Flutter learning pathway > State in Flutter apps > 12 Use ChangeNotifier to update app state > Test yourself
7 Review
8 Test yourself
State Management Quiz
1 / 2
What is a ChangeNotifier?
A. A built-in Dart class for sending push notifications.
B. A widget that displays notifications to the user.
C. A class that can notify listeners when its data changes, enabling reactive UI updates.
That's right!
ChangeNotifier provides the notifyListeners method to signal widgets to rebuild when state changes.
D. A type of animation controller in Flutter.
Previous question
Next question
Previous
Make Http Requests
Next
Use ListenableBuilder to update app UI

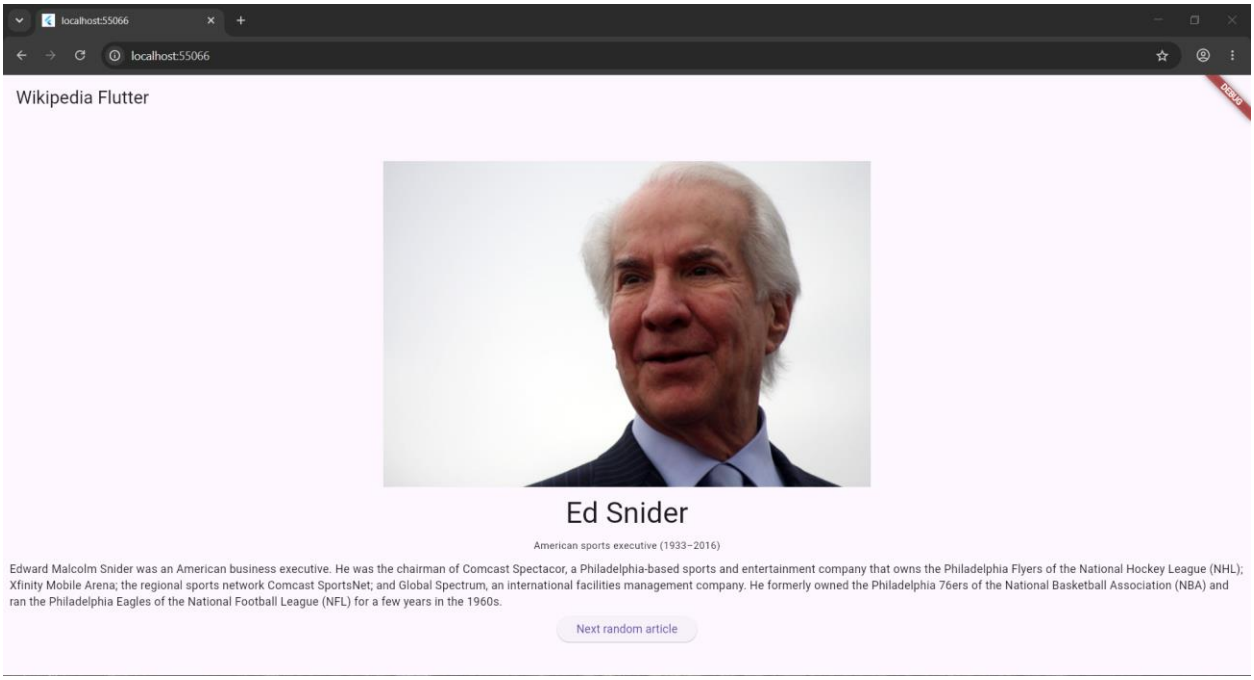


13/ Use ListenableBuilder to update app UI

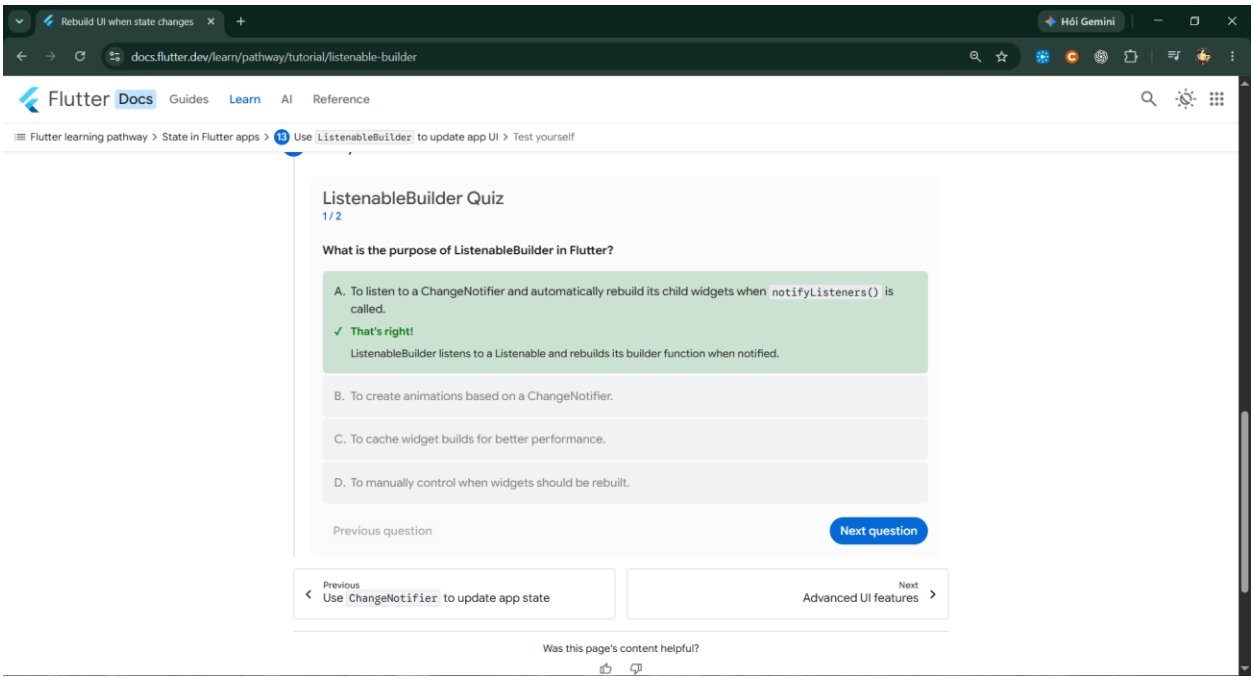
Code

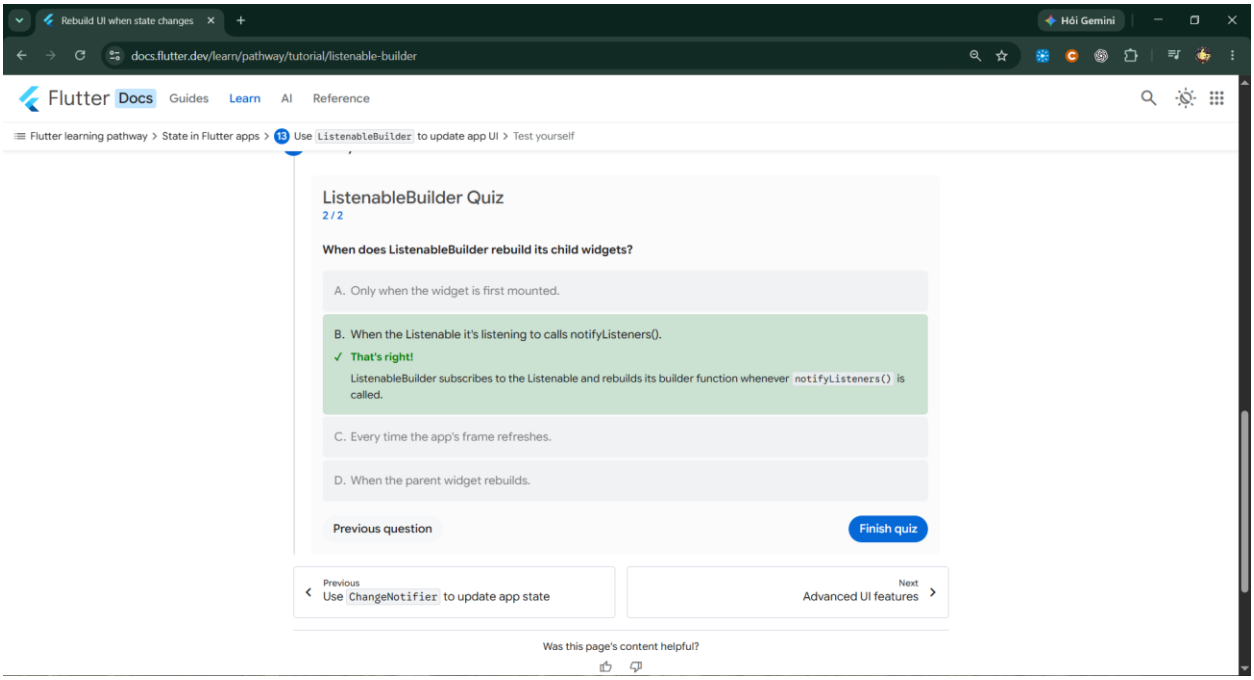


Output


localhost:55066
localhost:55066
Wikipedia Flutter
DEBUG
Ed Snider
American sports executive (1933–2016)
Edward Malcolm Snider was an American business executive. He was the chairman of Comcast Spectacor, a Philadelphia-based sports and entertainment company that owns the Philadelphia Flyers of the National Hockey League (NHL); Xfinity Mobile Arena; the regional sports network Comcast SportsNet; and Global Spectrum, an international facilities management company. He formerly owned the Philadelphia 76ers of the National Basketball Association (NBA) and ran the Philadelphia Eagles of the National Football League (NFL) for a few years in the 1960s.
Next random article

Quiz

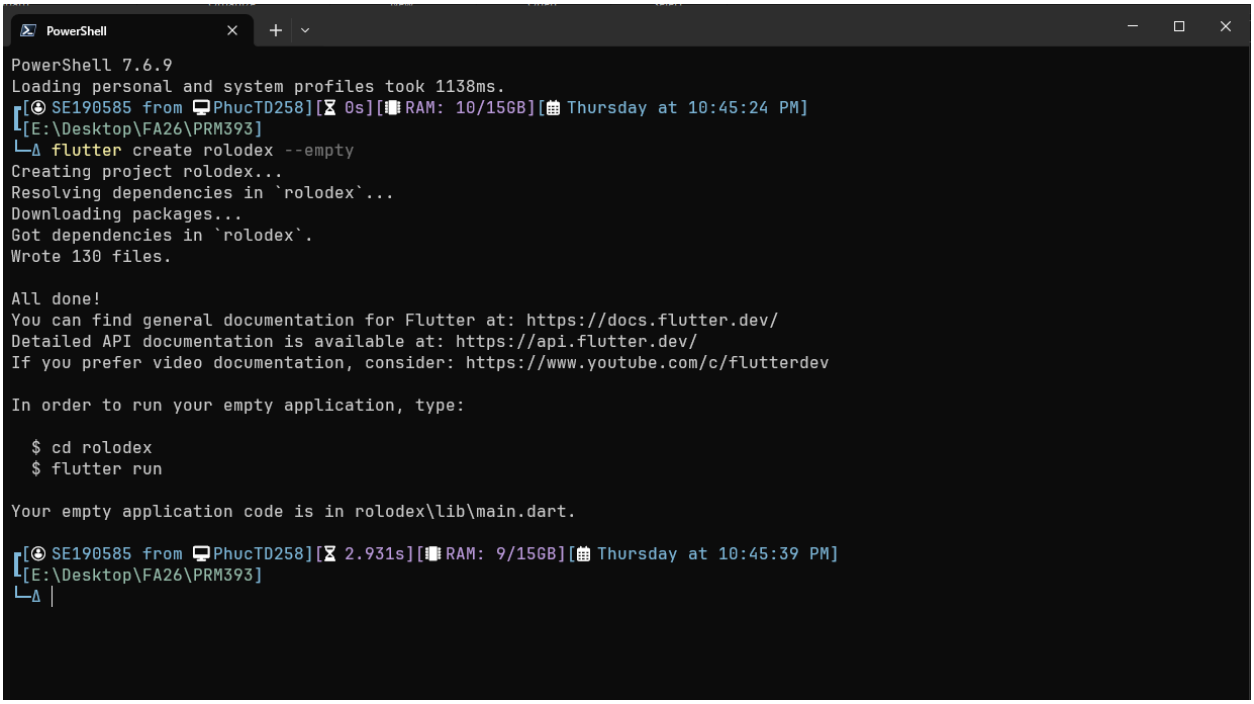

Rebuild UI when state changes
Hỏi Gemini
docs.flutter.dev/learn/pathway/tutorial/listenable-builder
Flutter Docs Guides Learn AI Reference
Flutter learning pathway > State in Flutter apps > 13 Use ListenableBuilder to update app UI > Test yourself
ListenableBuilder Quiz
1 / 2
What is the purpose of ListenableBuilder in Flutter?
A. To listen to a ChangeNotifier and automatically rebuild its child widgets when notifyListeners() is called.
✓ That's right!
ListenableBuilder listens to a Listenable and rebuilds its builder function when notified.
B. To create animations based on a ChangeNotifier.
C. To cache widget builds for better performance.
D. To manually control when widgets should be rebuilt.
Previous question
Next question
Previous
Use ChangeNotifier to update app state
Next
Advanced UI features
Was this page's content helpful?



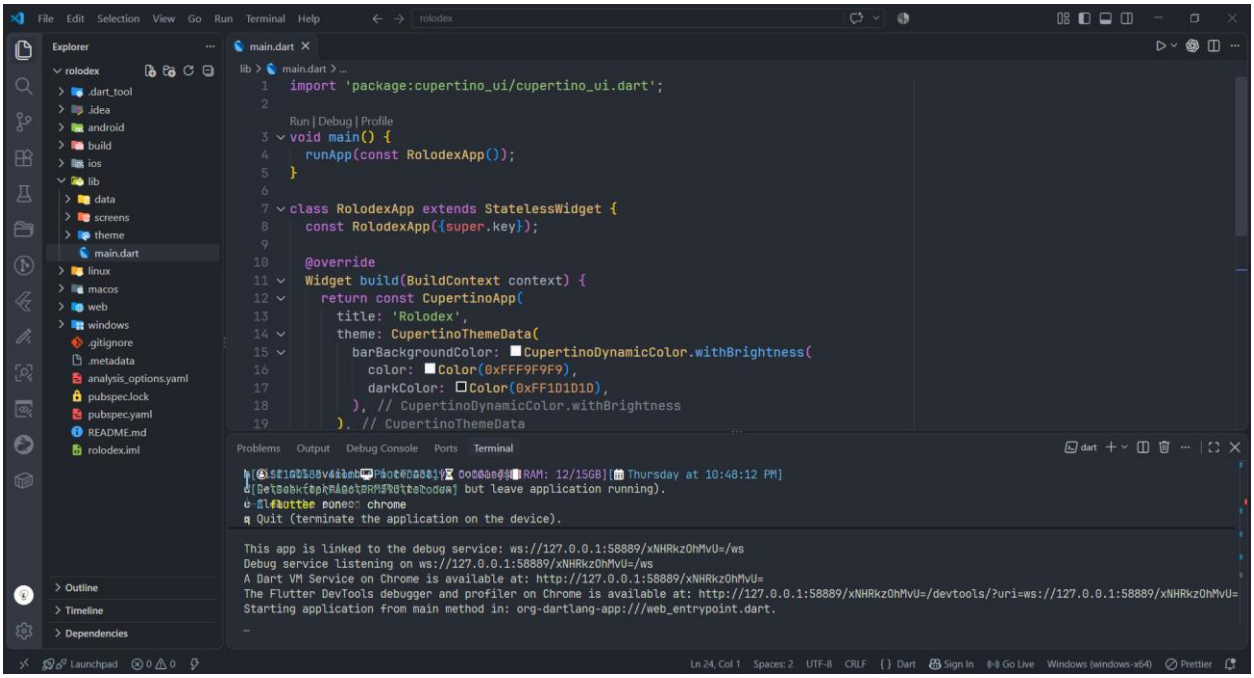
Project 3 – Rolodex (Flutter UI 102)

14/ Advanced UI features

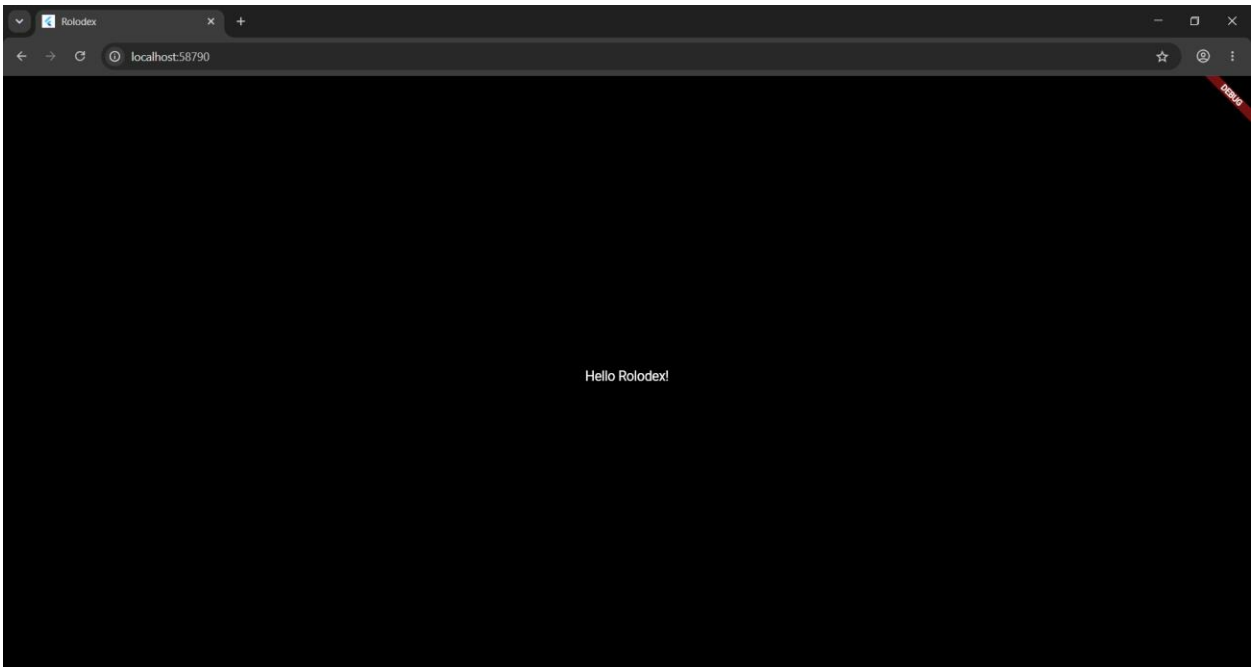
Terminal



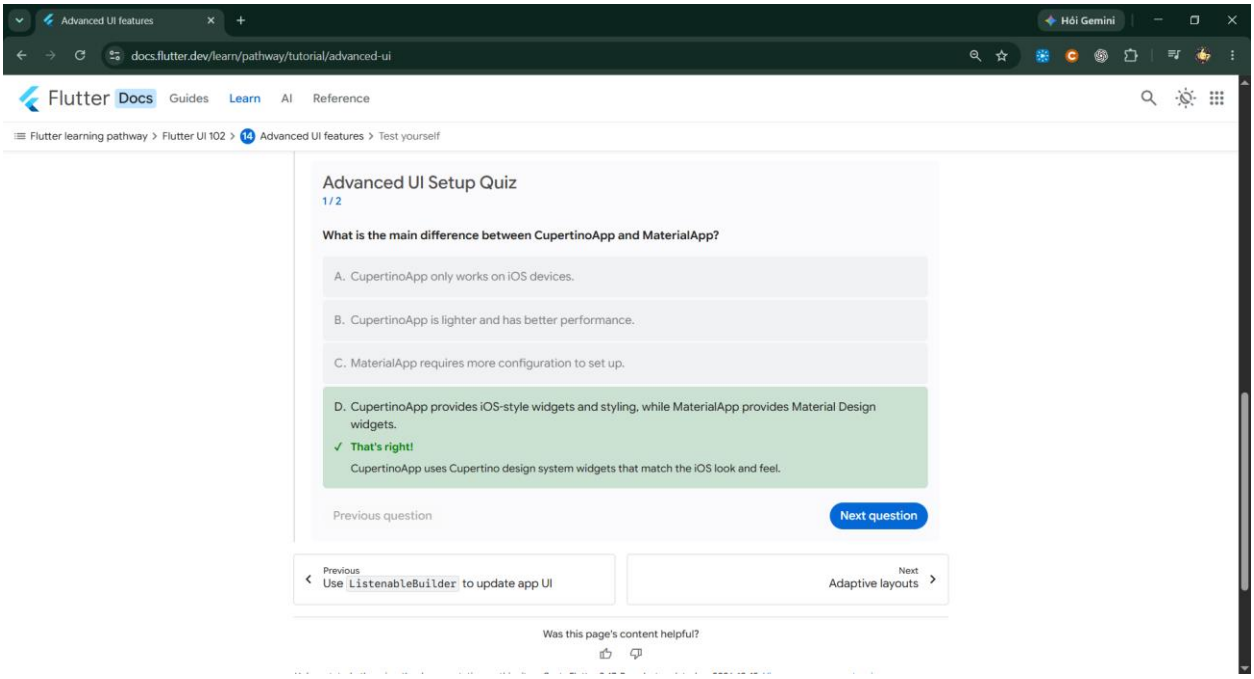
Code

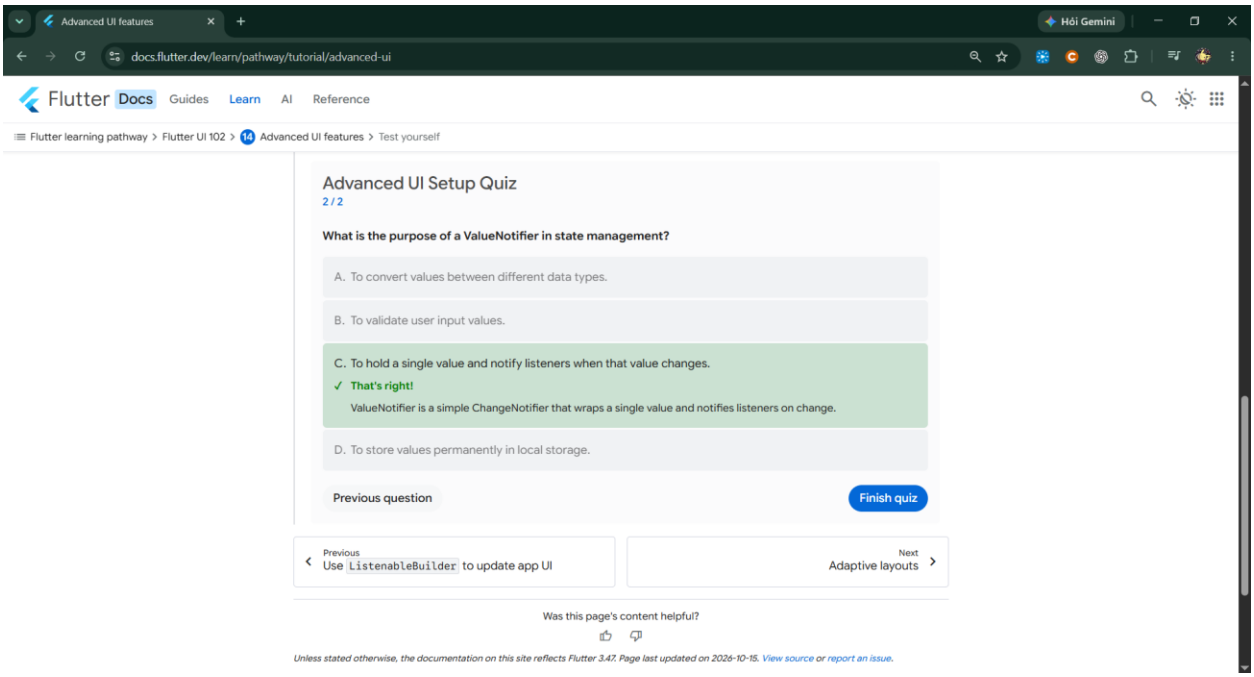

File Edit Selection View Go Run Terminal Help
rolodex
Explorer
rolodex
.dart_tool
.idea
android
build
ios
lib
data
screens
theme
main.dart
linux
macos
web
windows
.gitignore
.metadata
analysis_options.yaml
pubspec.lock
pubspec.yaml
README.md
rolodex.iml
Outline
Timeline
Dependencies
main.dart
lib > main.dart > ...
import 'package:cupertino_ui/cupertino_ui.dart';
Run | Debug | Profile
void main() {
runApp(const RolodexApp());
}
class RolodexApp extends StatelessWidget {
const RolodexApp({super.key});
@override
Widget build(BuildContext context) {
return const CupertinoApp(
title: 'Rolodex',
theme: CupertinoThemeData(
barBackgroundColor: CupertinoDynamicColor.withBrightness(
color: Color(0xFFF9F9F9),
darkColor: Color(0xFF1D1D1D),
), // CupertinoDynamicColor.withBrightness
), // CupertinoThemeData
Problems Output Debug Console Ports Terminal
dart
RAM: 12/15GB] Thursday at 10:48:12 PM]
but leave application running).
flutter run chrome
Quit (terminate the application on the device).
This app is linked to the debug service: ws://127.0.0.1:58889/xNHRkzOhMvU=/ws
Debug service listening on ws://127.0.0.1:58889/xNHRkzOhMvU=/ws
A Dart VM Service on Chrome is available at: http://127.0.0.1:58889/xNHRkzOhMvU=
The Flutter DevTools debugger and profiler on Chrome is available at: http://127.0.0.1:58889/xNHRkzOhMvU=/devtools/?uri=ws://127.0.0.1:58889/xNHRkzOhMvU=
Starting application from main method in: org-dartlang-app:///web_entrypoint.dart.
Launchpad
Ln 24, Col 1
Spaces: 2
UTF-8
CRLF
Dart
Sign In
Go Live
Windows (windows-x64)
Prettier

Output


Rolodex
localhost:58790
DEBUG
Hello Rolodex!

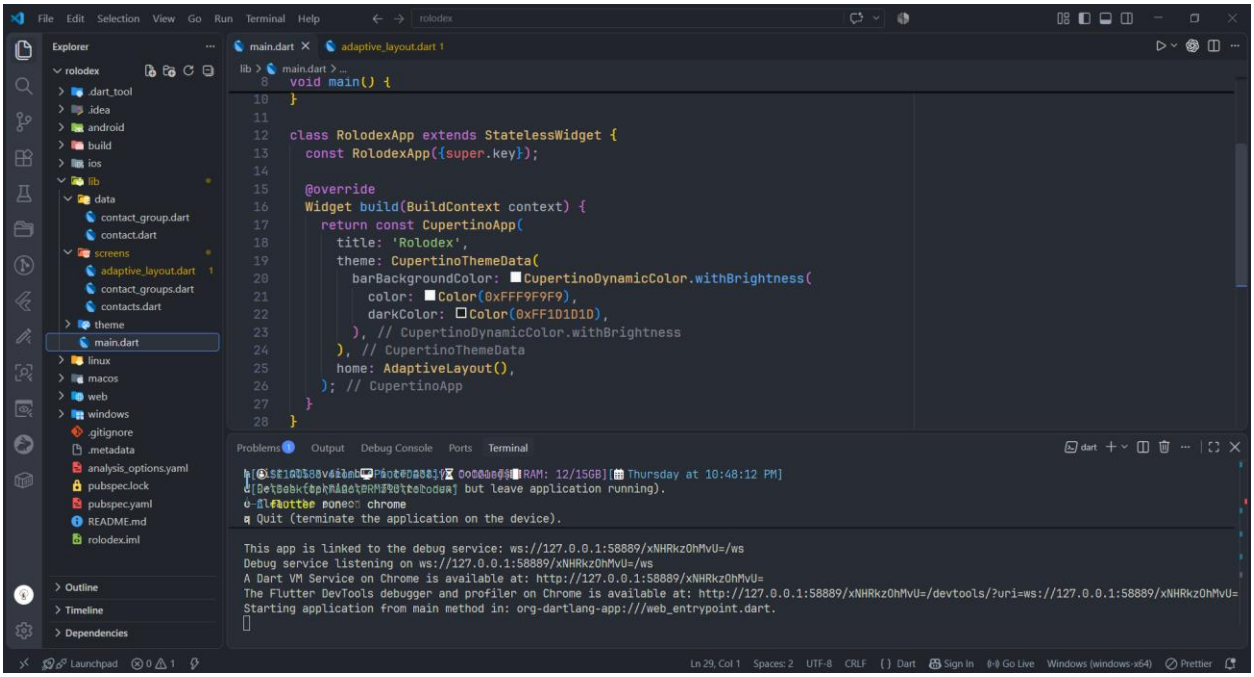
Quiz

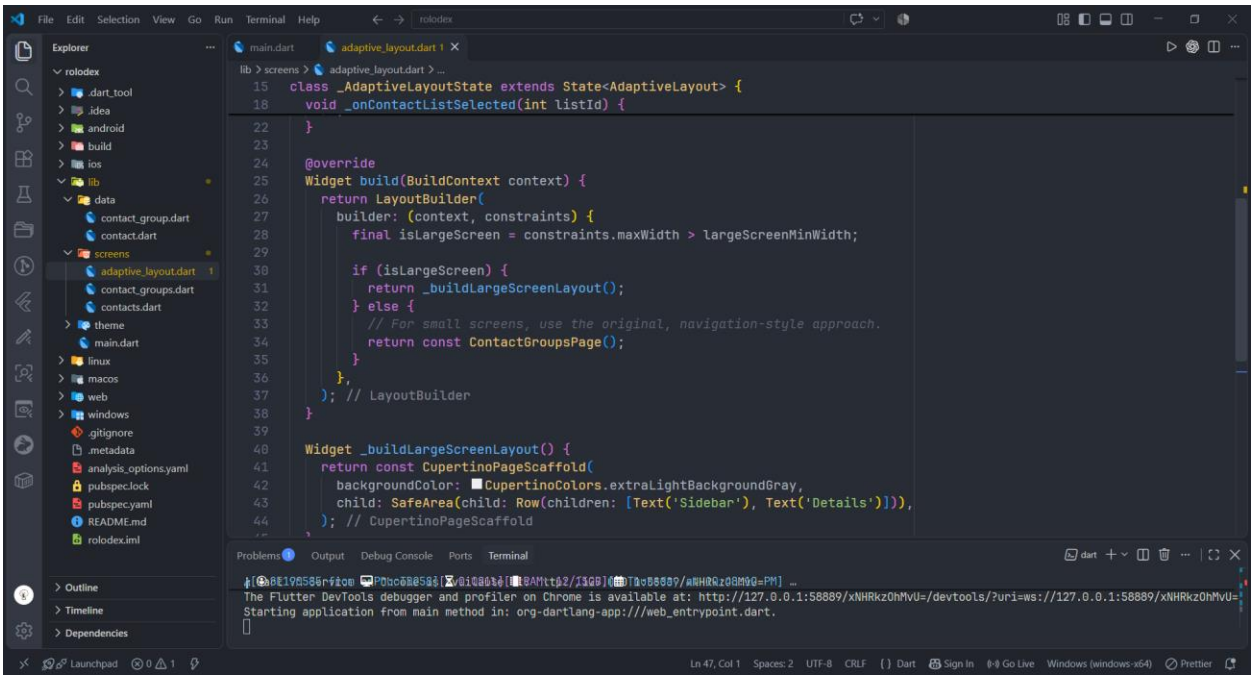




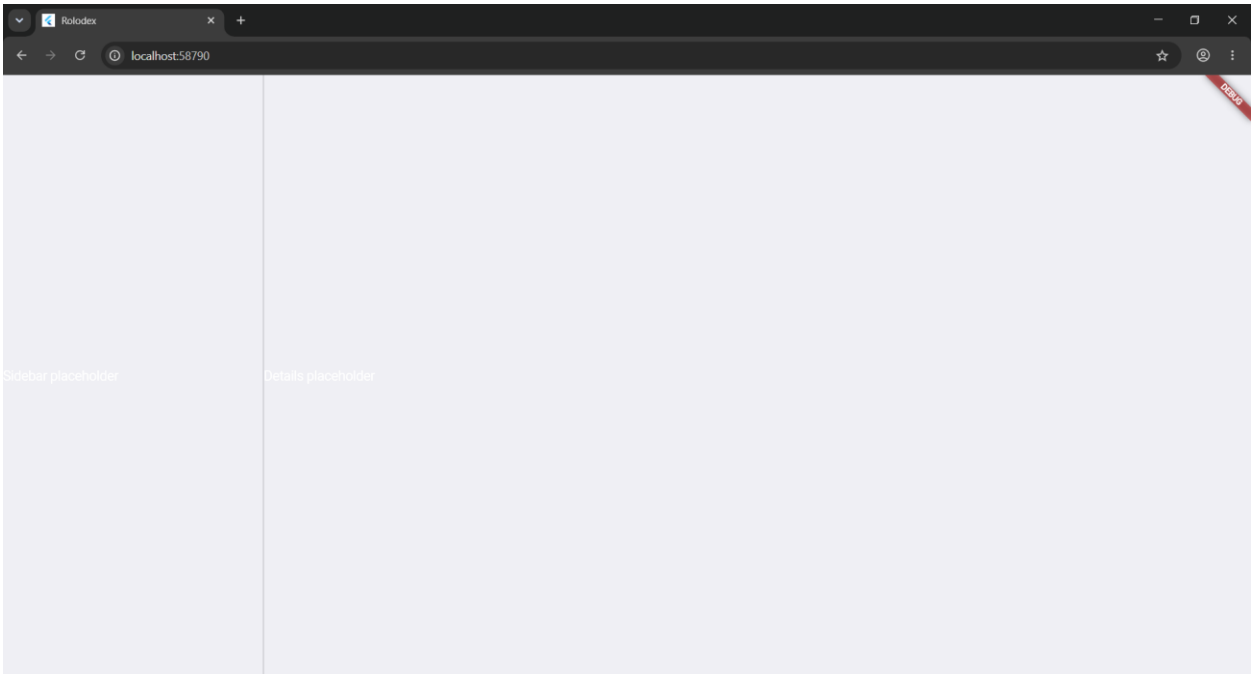
15/ Adaptive layouts

Code

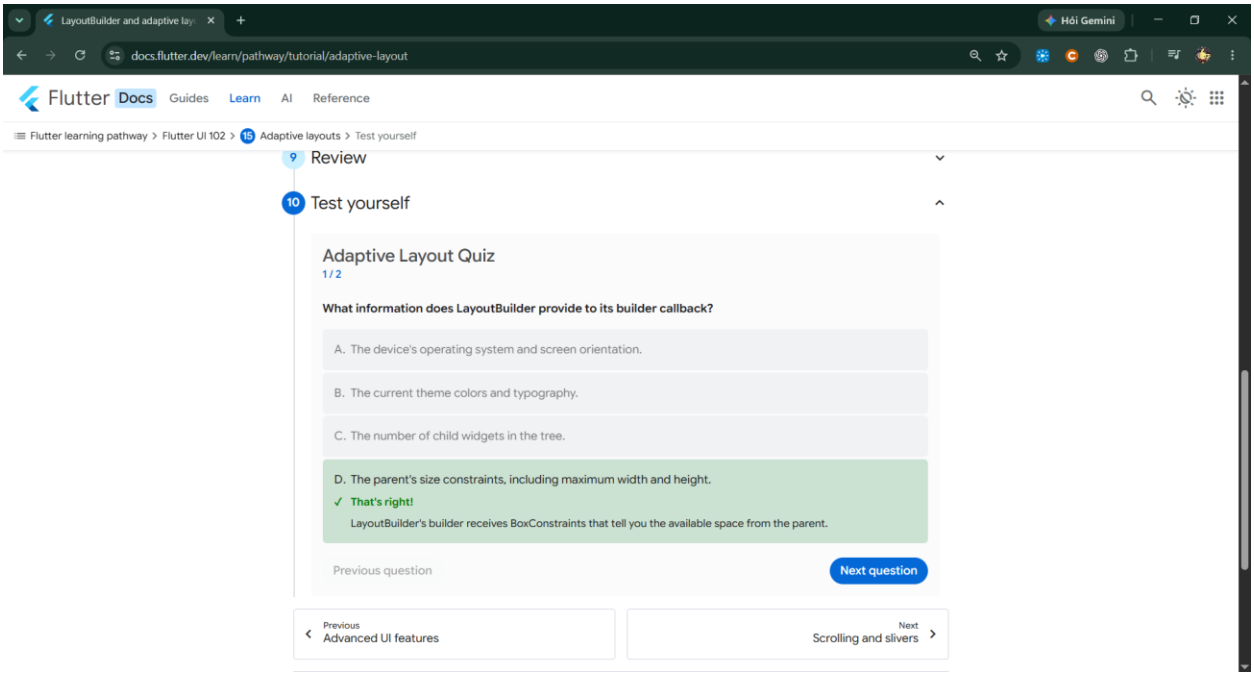

File Edit Selection View Go Run Terminal Help
rolodex
Explorer
main.dart
adaptive_layout.dart 1
lib > main.dart > ...
void main() {
}
class RolodexApp extends StatelessWidget {
const RolodexApp({super.key});
@override
Widget build(BuildContext context) {
return const CupertinoApp(
title: 'Rolodex',
theme: CupertinoThemeData(
barBackgroundColor: CupertinoDynamicColor.withBrightness(
color: Color(0xFFF9F9F9),
darkColor: Color(0xFF1D1D1D),
), // CupertinoDynamicColor.withBrightness
), // CupertinoThemeData
home: AdaptiveLayout(),
); // CupertinoApp
}
}
Problems
Output
Debug Console
Ports
Terminal
Thursday at 10:48:12 PM]
Quit (terminate the application on the device).
This app is linked to the debug service: ws://127.0.0.1:58889/xNHRkzOhMvU=/ws
Debug service listening on ws://127.0.0.1:58889/xNHRkzOhMvU=/ws
A Dart VM Service on Chrome is available at: http://127.0.0.1:58889/xNHRkzOhMvU=
The Flutter DevTools debugger and profiler on Chrome is available at: http://127.0.0.1:58889/xNHRkzOhMvU=/devtools/?uri=ws://127.0.0.1:58889/xNHRkzOhMvU=
Starting application from main method in: org-dartlang-app:///web_entrypoint.dart.
Outline
Timeline
Dependencies
Launchpad
Ln 29, Col 1
Spaces: 2
UTF-8
CRLF
Dart
Sign In
Go Live
Windows (windows-x64)
Prettier

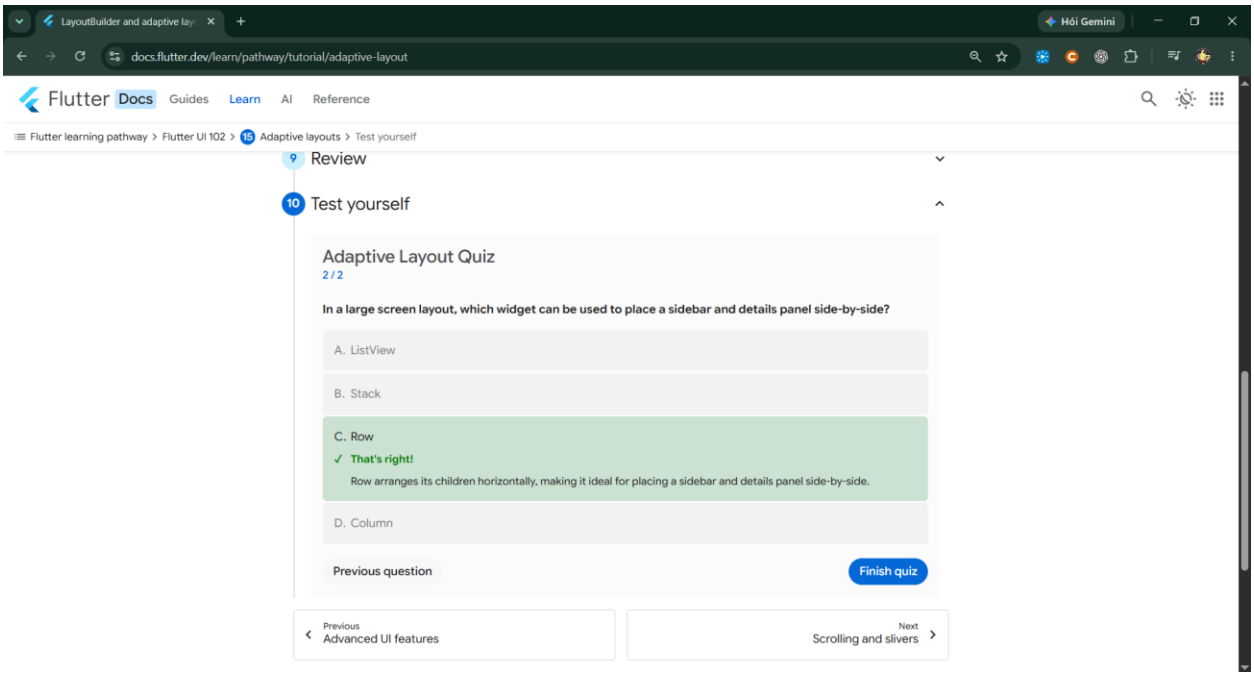

File Edit Selection View Go Run Terminal Help
rolodex
Explorer
main.dart
adaptive_layout.dart 1
lib > screens > adaptive_layout.dart > ...
class _AdaptiveLayoutState extends State<AdaptiveLayout> {
void _onContactListSelected(int listId) {
}
@override
Widget build(BuildContext context) {
return LayoutBuilder(
builder: (context, constraints) {
final isLargeScreen = constraints.maxWidth > largeScreenMinWidth;
if (isLargeScreen) {
return _buildLargeScreenLayout();
} else {
// For small screens, use the original, navigation-style approach.
return const ContactGroupsPage();
}
},
); // LayoutBuilder
}
Widget _buildLargeScreenLayout() {
return const CupertinoPageScaffold(
backgroundColor: CupertinoColors.extraLightBackgroundGray,
child: SafeArea(child: Row(children: [Text('Sidebar'), Text('Details')])),
); // CupertinoPageScaffold
Problems
Output
Debug Console
Ports
Terminal
The Flutter DevTools debugger and profiler on Chrome is available at: http://127.0.0.1:58889/xNHRkzOhMvU=/devtools/?uri=ws://127.0.0.1:58889/xNHRkzOhMvU=
Starting application from main method in: org-dartlang-app:///web_entrypoint.dart.
Outline
Timeline
Dependencies
Launchpad
Ln 47, Col 1
Spaces: 2
UTF-8
CRLF
Dart
Sign In
Go Live
Windows (windows-x64)
Prettier

Output


Rolodex
localhost:58790
DEBUG
Sidebar placeholder
Details placeholder

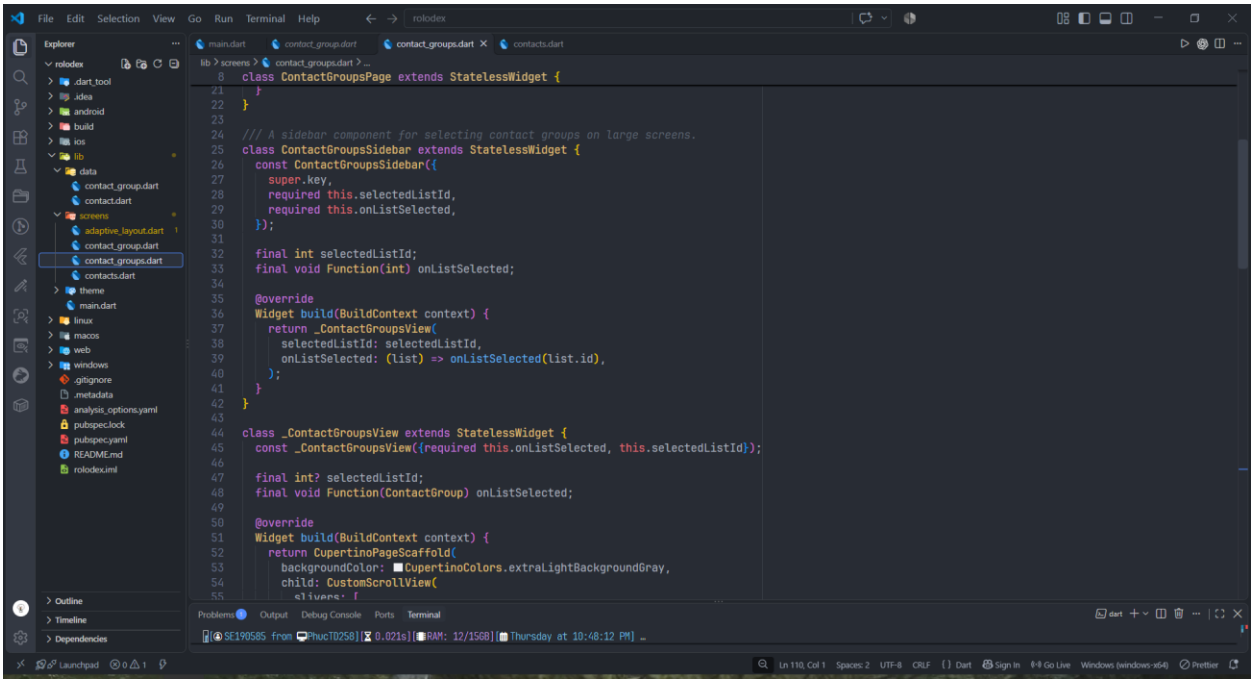
Quiz


LayoutBuilder and adaptive lay
Hỏi Gemini
docs.flutter.dev/learn/pathway/tutorial/adaptive-layout
Flutter Docs Guides Learn AI Reference
Flutter learning pathway > Flutter UI 102 > 15 Adaptive layouts > Test yourself
9 Review
10 Test yourself
Adaptive Layout Quiz
1 / 2
What information does LayoutBuilder provide to its builder callback?
A. The device's operating system and screen orientation.
B. The current theme colors and typography.
C. The number of child widgets in the tree.
D. The parent's size constraints, including maximum width and height.
That's right!
LayoutBuilder's builder receives BoxConstraints that tell you the available space from the parent.
Previous question
Next question
Previous
Advanced UI features
Next
Scrolling and slivers

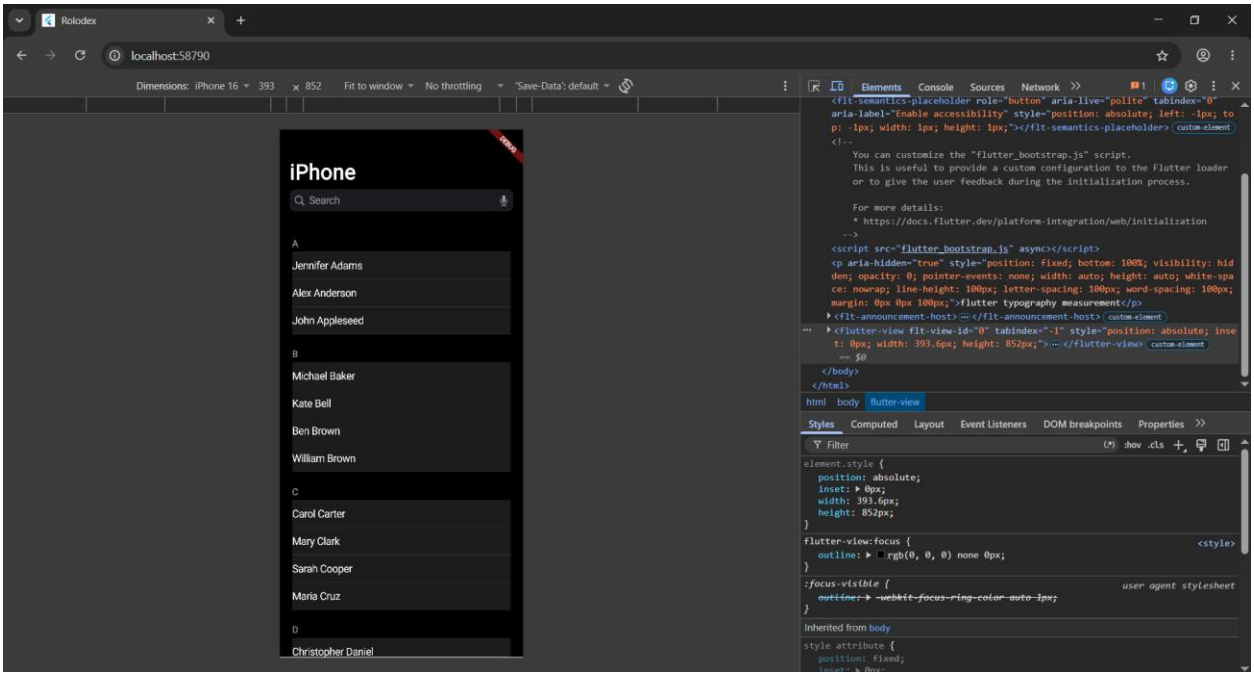


16/ Scrolling and slivers

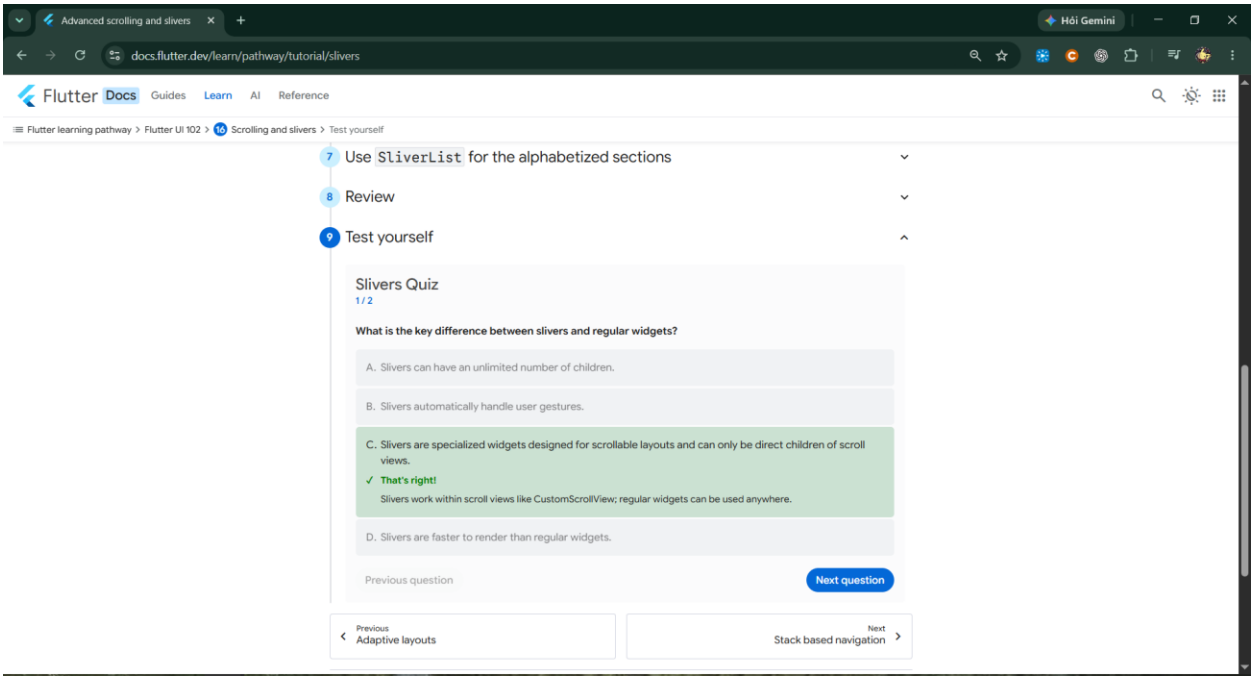
Code

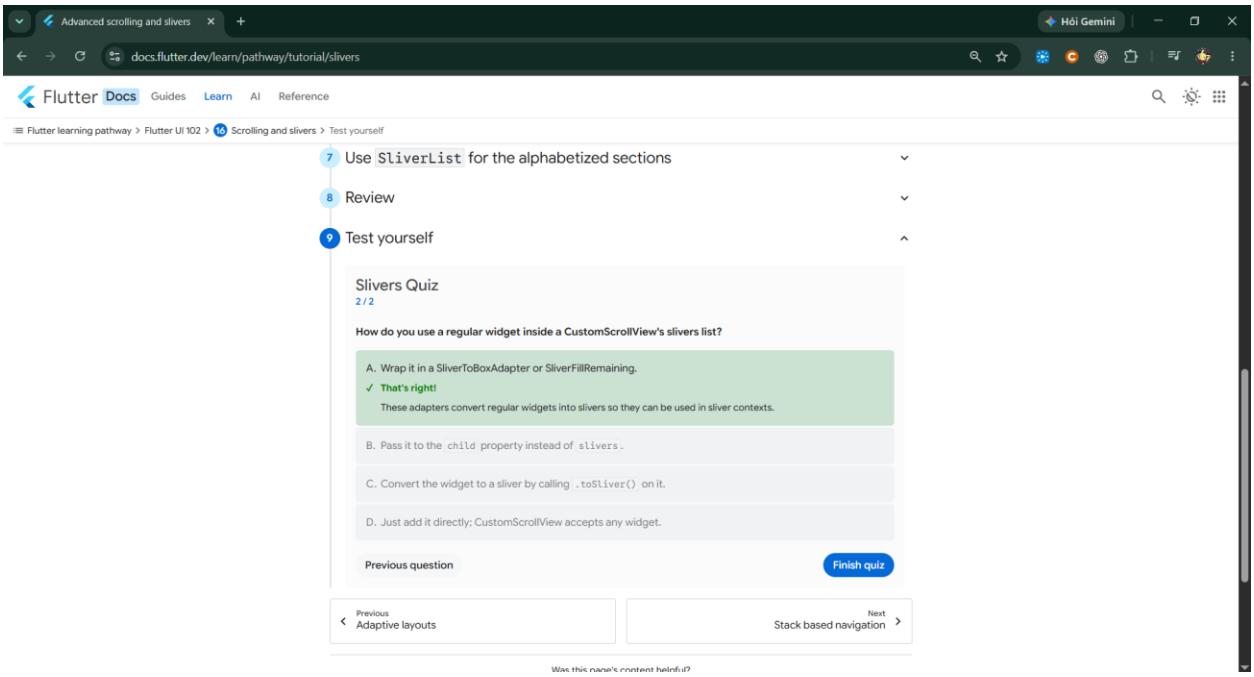


Output


Rolodex
localhost:58790
iPhone
Search
A
Jennifer Adams
Alex Anderson
John Appleseed
B
Michael Baker
Kate Bell
Ben Brown
William Brown
C
Carol Carter
Mary Clark
Sarah Cooper
Maria Cruz
D
Christopher Daniel
Elements
Console
Sources
Network

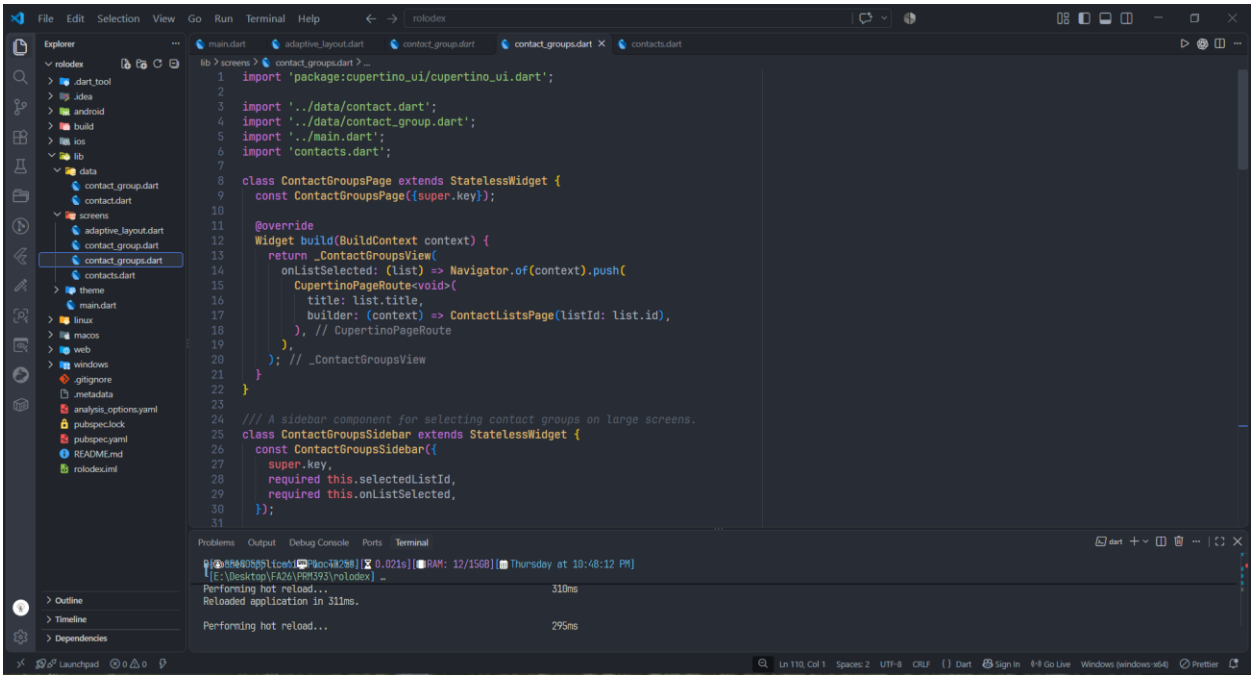
Quiz


Advanced scrolling and slivers
docs.flutter.dev/learn/pathway/tutorial/slivers
Flutter Docs
Guides
Learn
AI
Reference
Use SliverList for the alphabetized sections
Review
Test yourself
Slivers Quiz
1 / 2
What is the key difference between slivers and regular widgets?
A. Slivers can have an unlimited number of children.
B. Slivers automatically handle user gestures.
C. Slivers are specialized widgets designed for scrollable layouts and can only be direct children of scroll views.
That's right!
Slivers work within scroll views like CustomScrollView; regular widgets can be used anywhere.
D. Slivers are faster to render than regular widgets.
Previous question
Next question
Previous
Adaptive layouts
Next
Stack based navigation

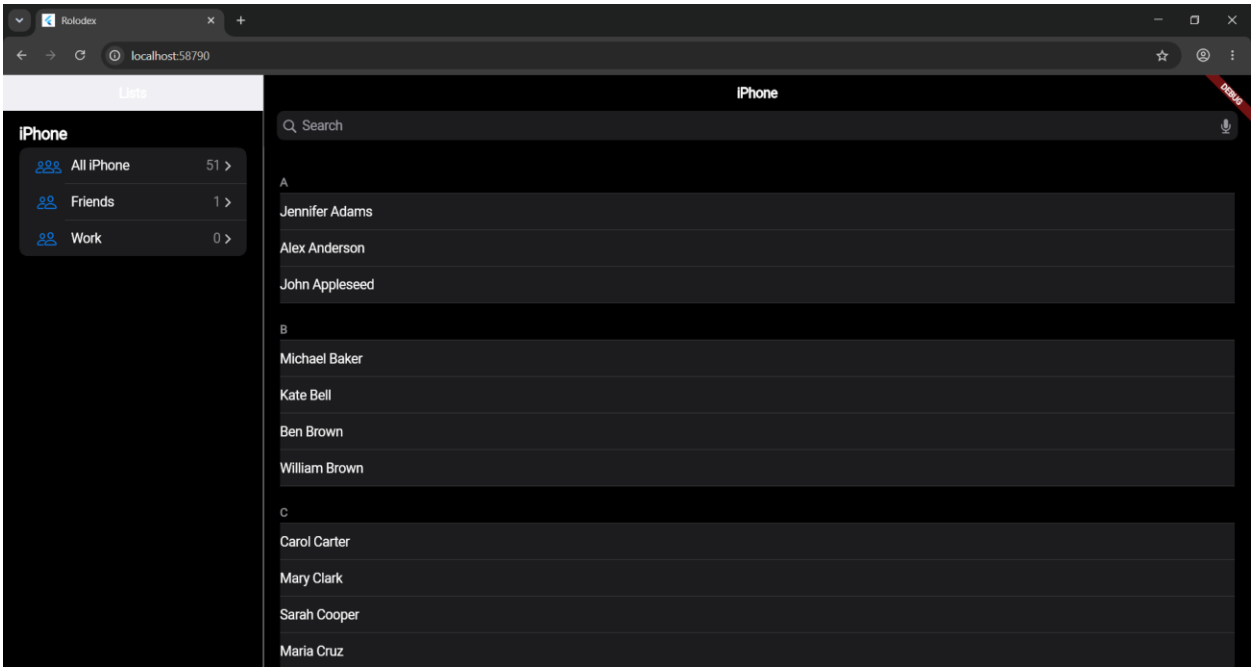


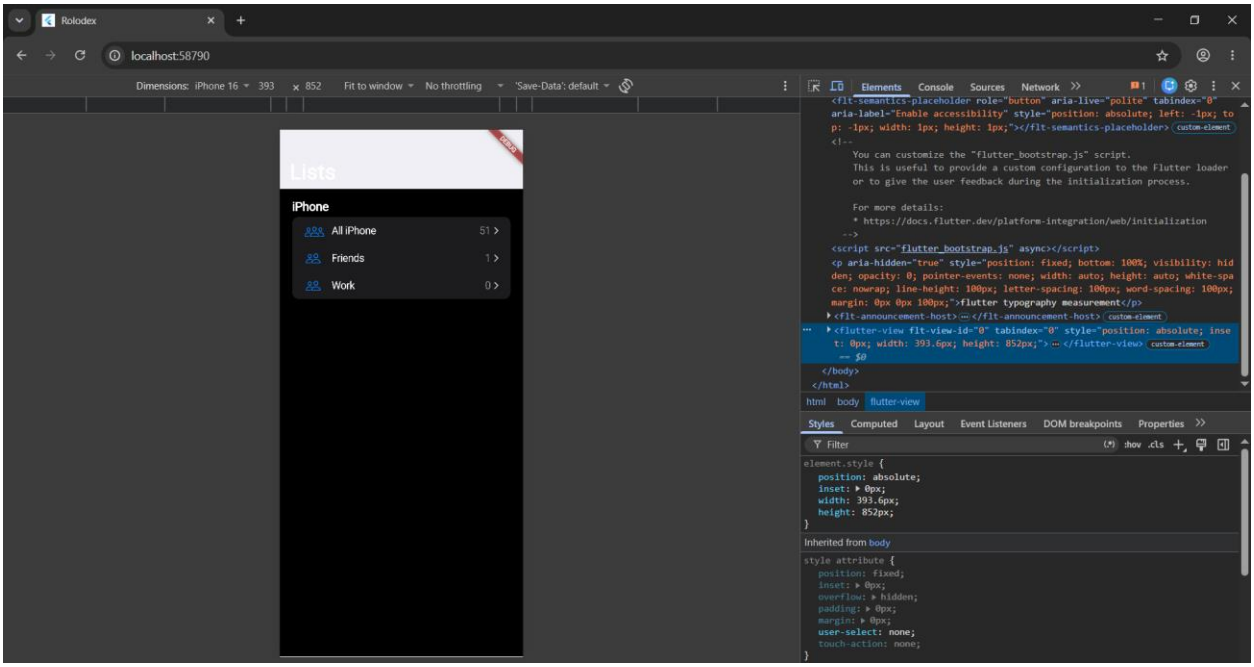
17/ Stack based navigation

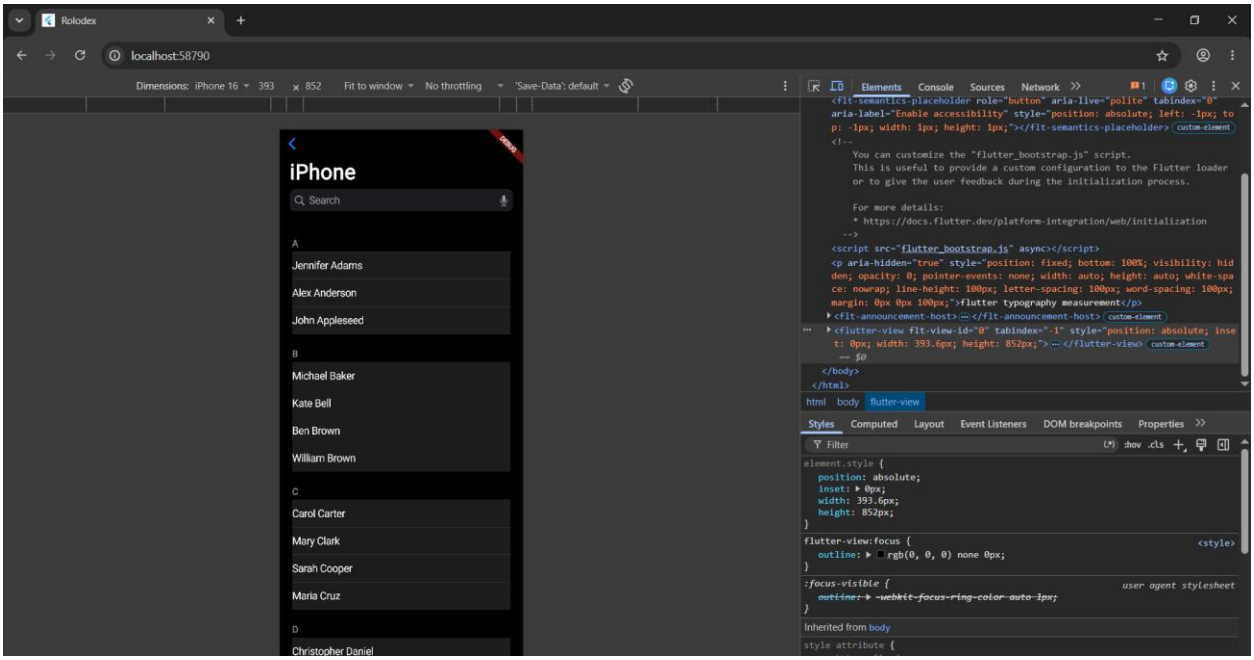
Code

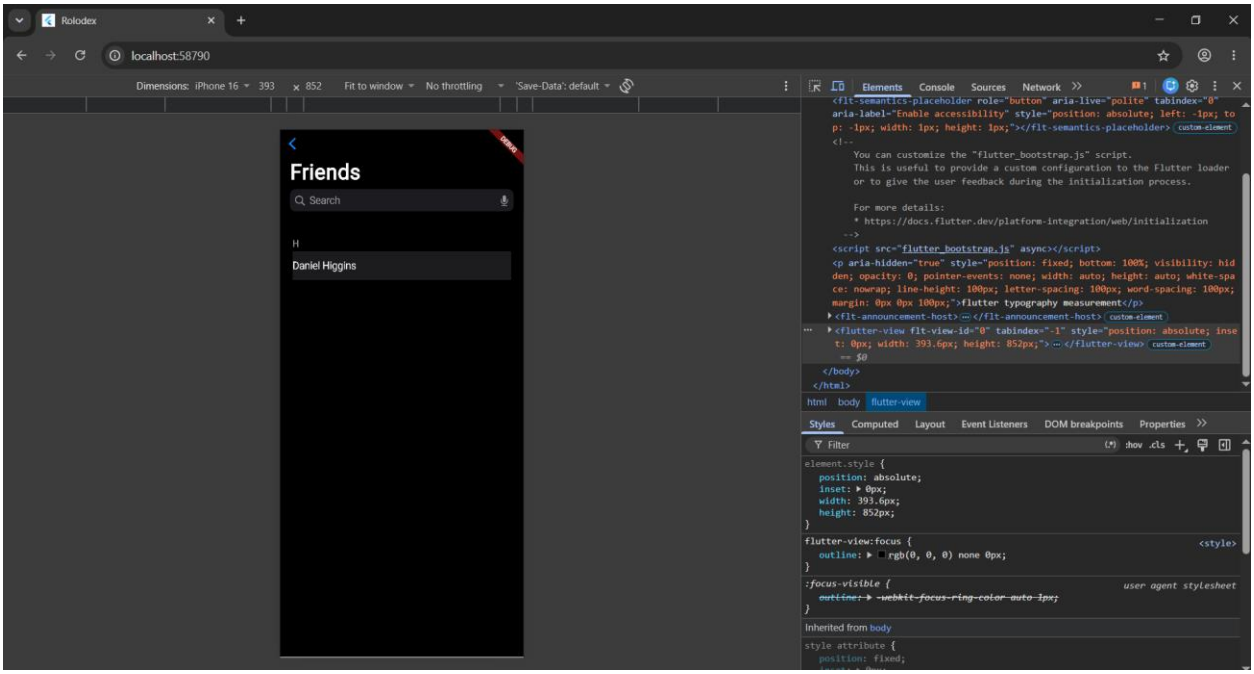


Output

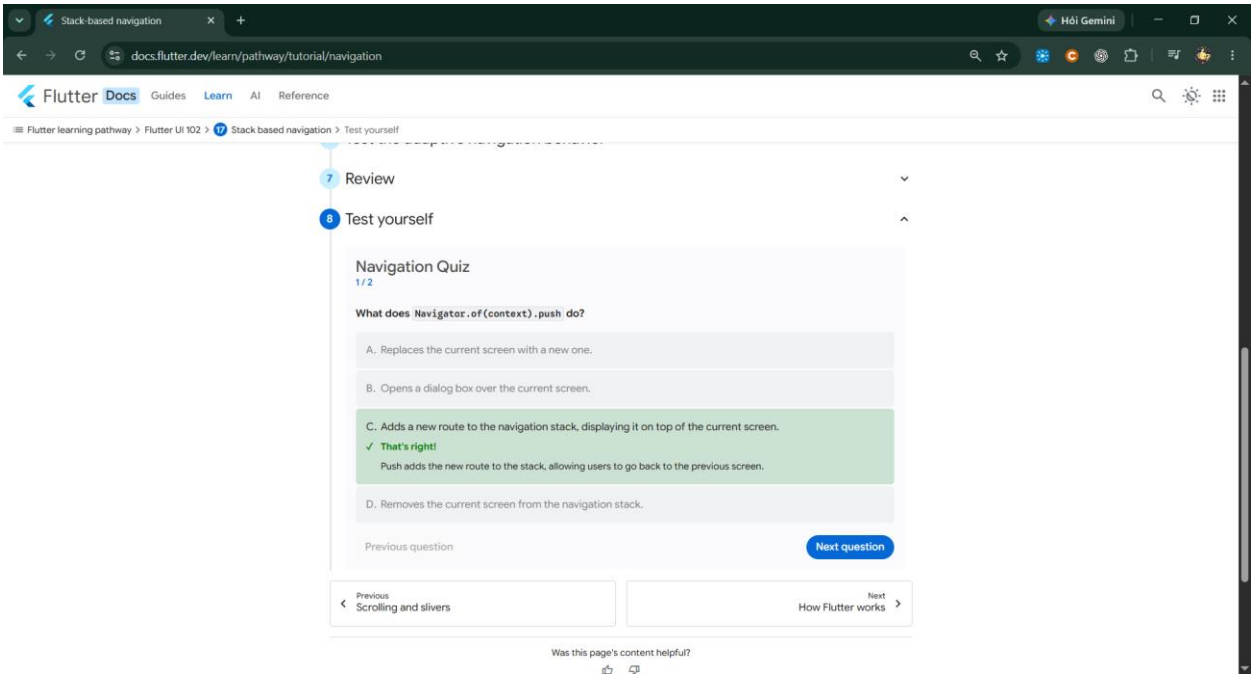

Rolodex
localhost:58790
Lists
iPhone
All iPhone 51
Friends 1
Work 0
iPhone
Search
DEBUG
A
Jennifer Adams
Alex Anderson
John Appleseed
B
Michael Baker
Kate Bell
Ben Brown
William Brown
C
Carol Carter
Mary Clark
Sarah Cooper
Maria Cruz

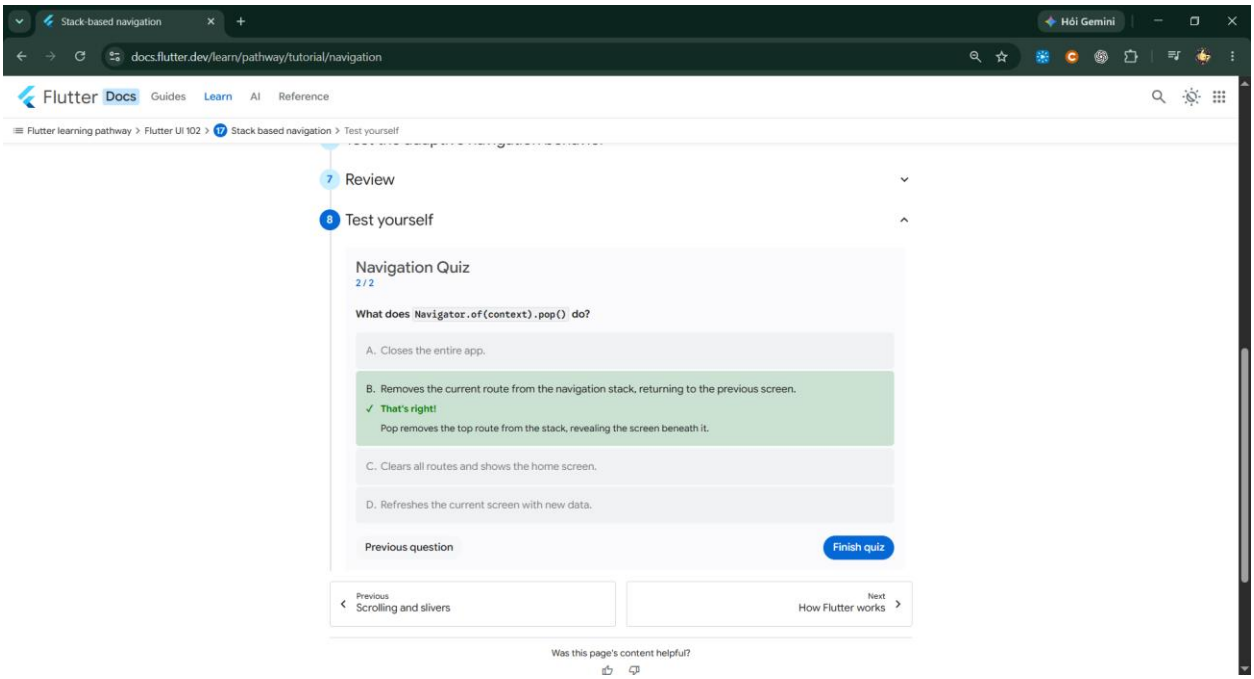

Rolodex
localhost:58790
Dimensions: iPhone 16 393 x 852 Fit to window No throttling 'Save-Data': default
Elements Console Sources Network
Lists
iPhone
All iPhone 51
Friends 1
Work 0
Styles Computed Layout Event Listeners DOM breakpoints Properties
element.style {
position: absolute;
inset: 0px;
width: 393.6px;
height: 852px;
}
Inherited from body
style attribute {
position: fixed;
inset: 0px;
overflow: hidden;
padding: 0px;
margin: 0px;
user-select: none;
touch-action: none;
}


Rolodex
localhost:58790
iPhone
Search
A
Jennifer Adams
Alex Anderson
John Appleseed
B
Michael Baker
Kate Bell
Ben Brown
William Brown
C
Carol Carter
Mary Clark
Sarah Cooper
Maria Cruz
D
Christopher Daniel

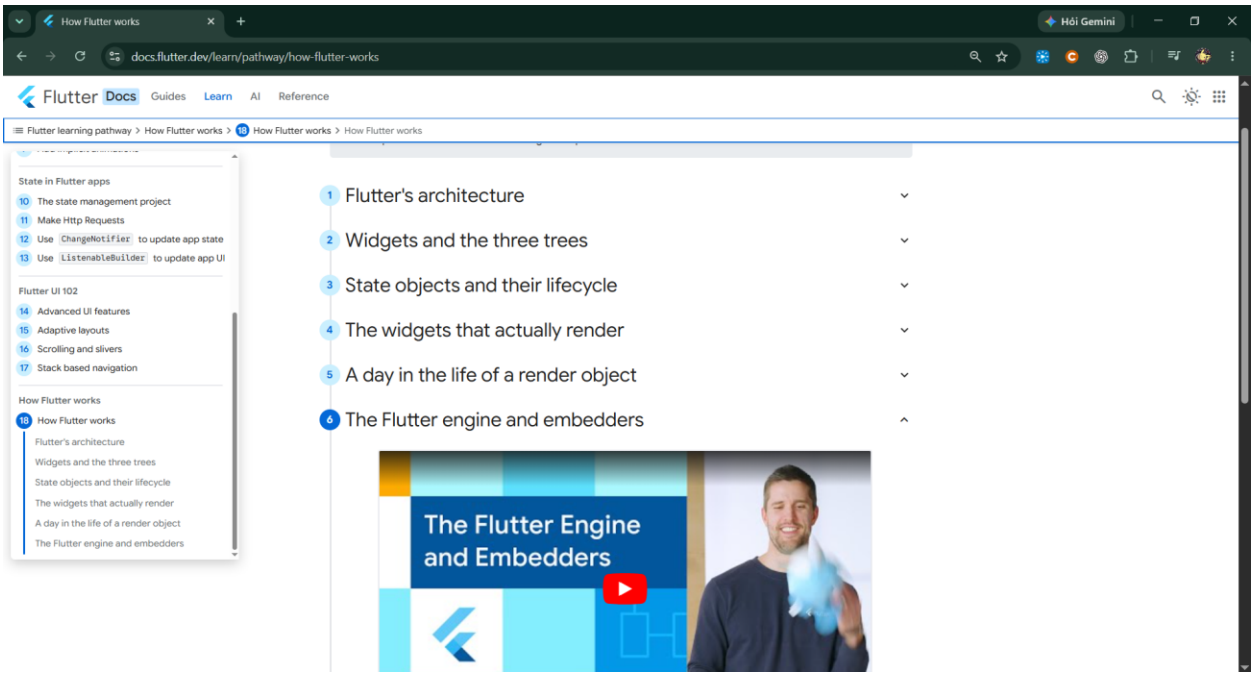

Rolodex
localhost:58790
Friends
Search
H
Daniel Higgins

Quiz





18/ How Flutter works



References link

- Flutter Learning Pathway: https://docs.flutter.dev/learn/pathway
- Link Github: https://github.com/phuc258/FlutterCourseEvidence.git